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(54) **MEMORY COMPACTION MANAGEMENT IN MEMORY DEVICES**

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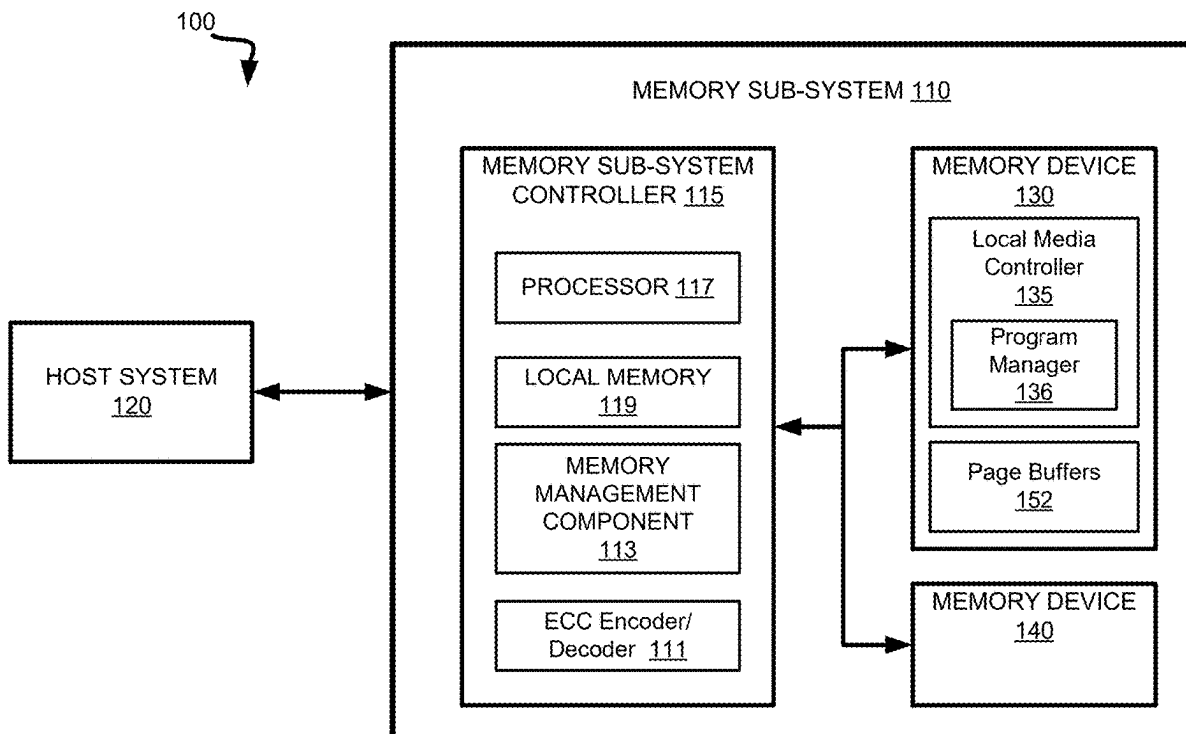
(63) Continuation of application No. 18/432,326, filed on Feb. 5, 2024, now Pat. No. 12,332,742, which is a continuation of application No. 17/859,468, filed on Jul. 7, 2022, now Pat. No. 11,934,266.

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(57)

ABSTRACT

Systems and methods are disclosed including a memory device and a processing device operatively coupled to the memory device. The processing device can perform operations comprising performing a data integrity check on a first set of memory cells to obtain a data integrity metric value; responsive to determining that the data integrity metric value satisfies a threshold criterion, determining whether a threshold number of second sets of memory cells, configured to store the first number of bits per memory cell, are available for a memory management operation; and responsive to determining that the threshold number of second sets of memory cells are available, causing the memory device to perform the memory management operation by copying data from the first set of memory cells and from the threshold number of second sets of memory cells to a third set of memory cells.



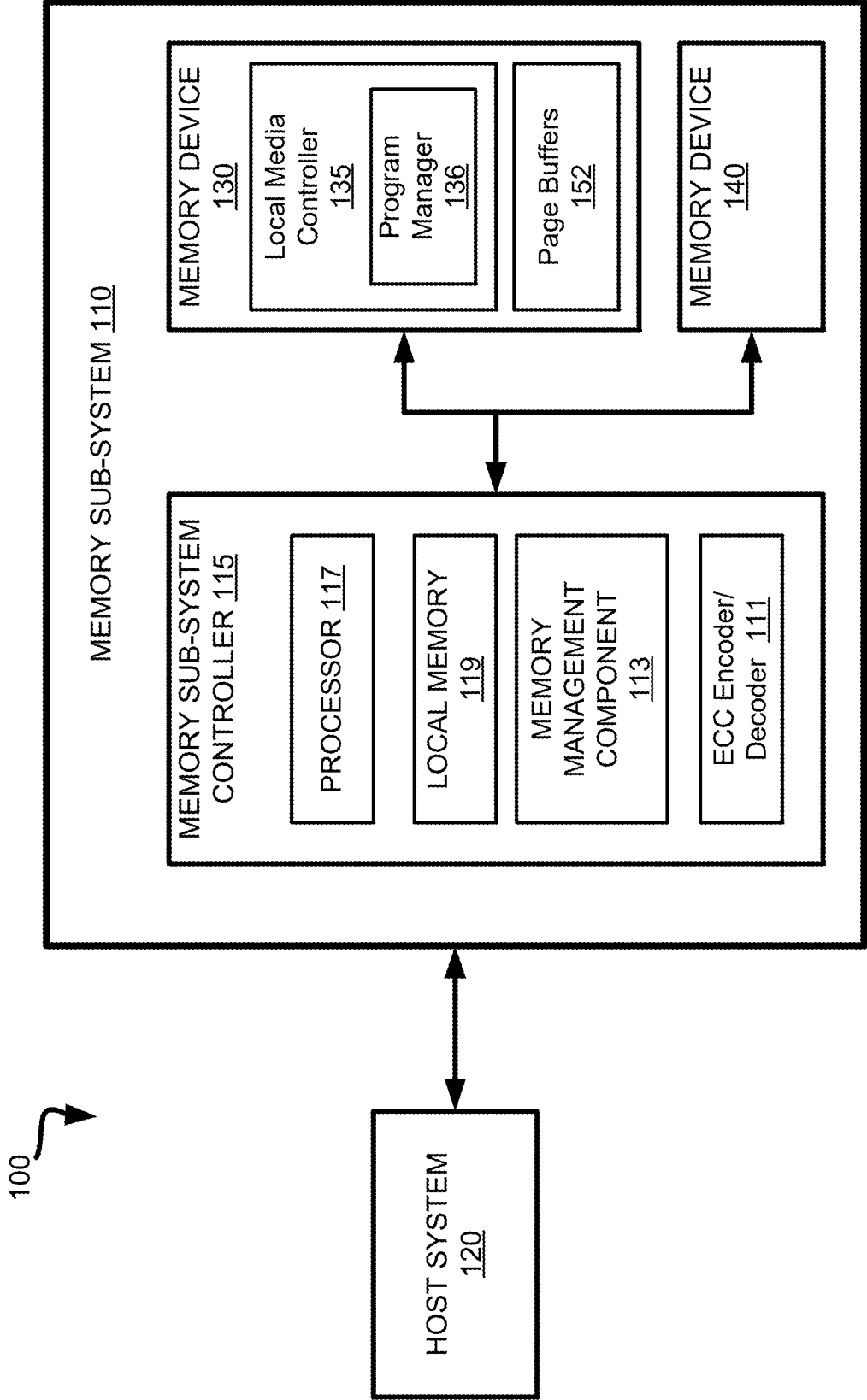


FIG. 1A

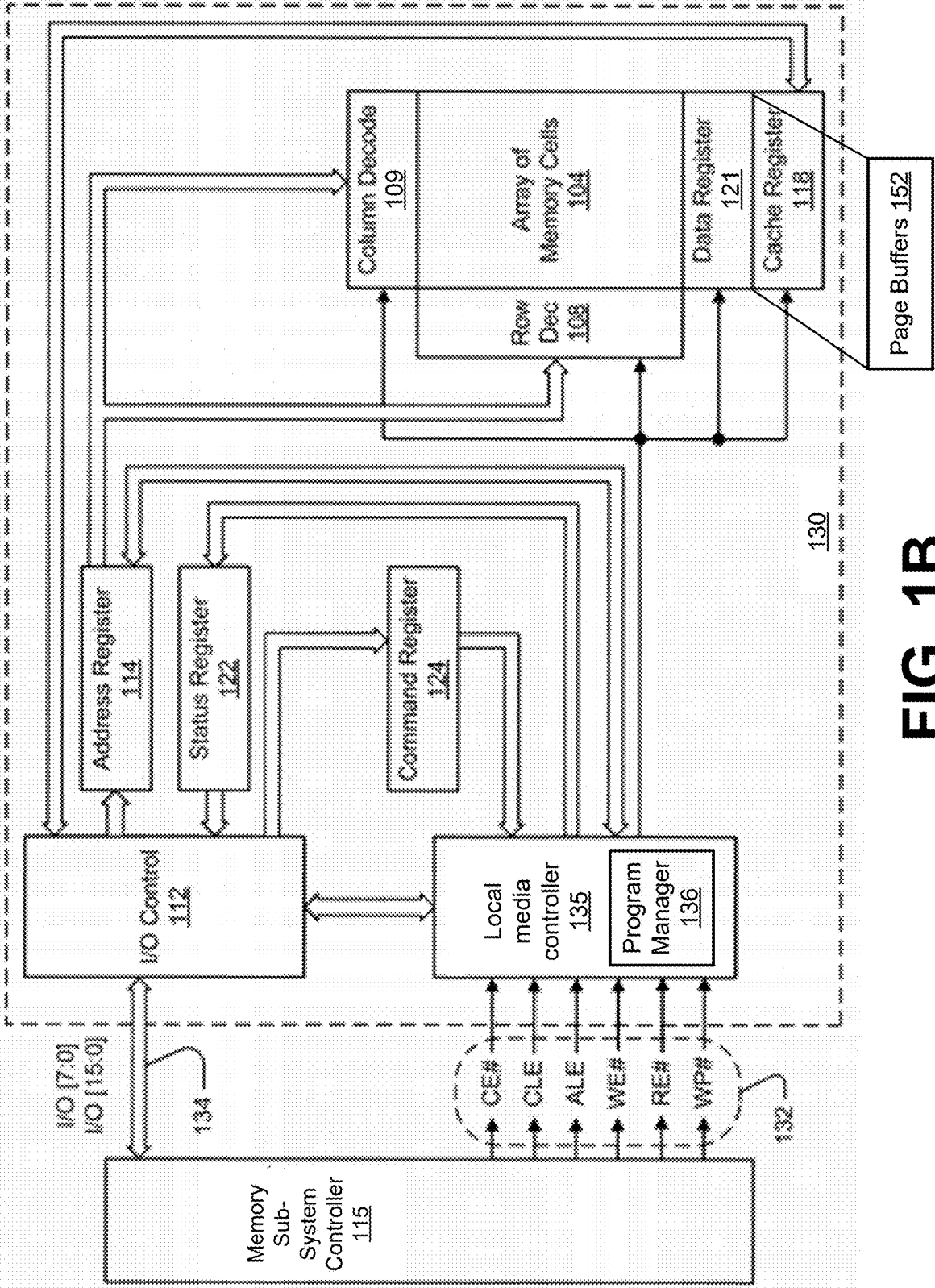


FIG. 1B

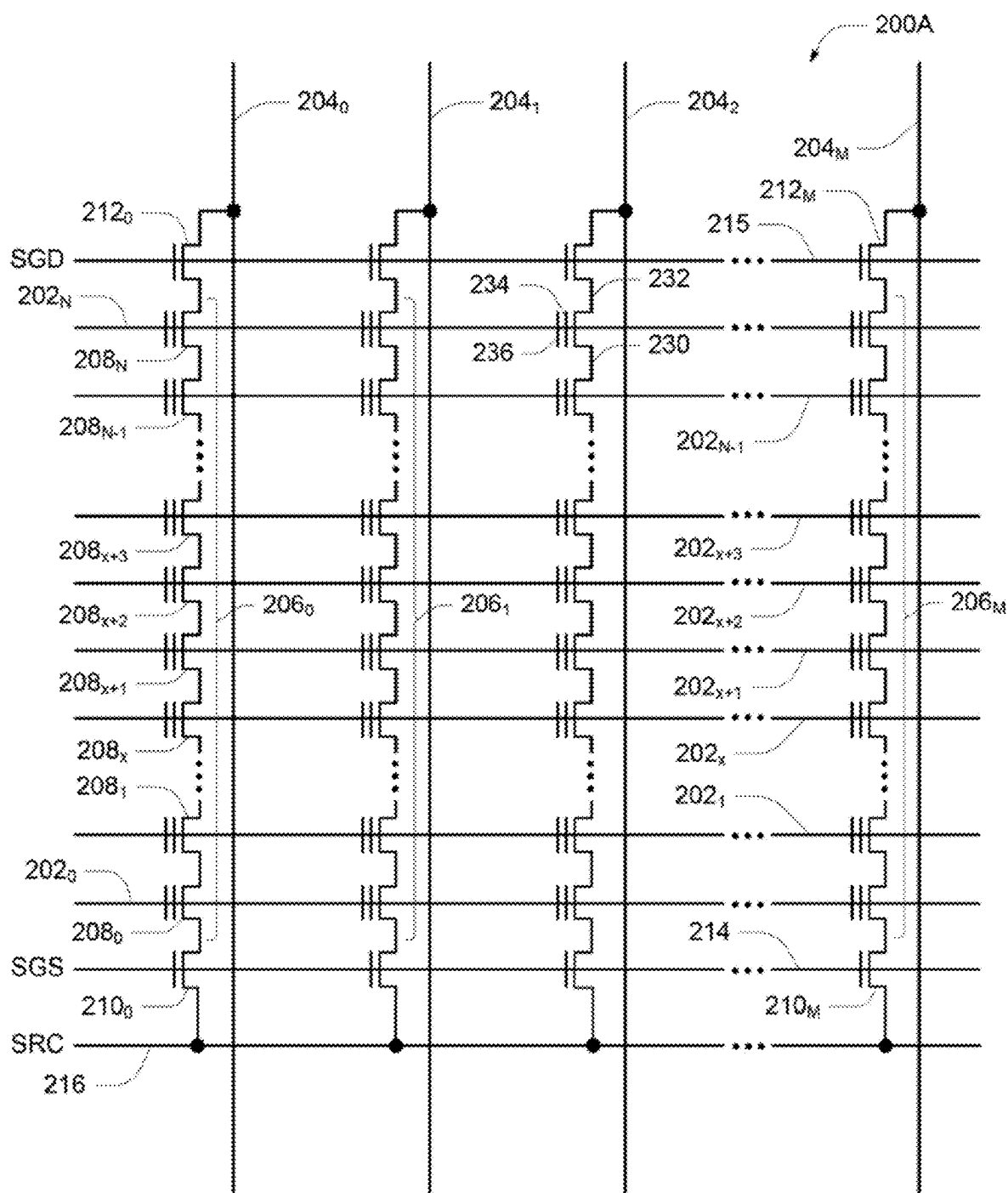


FIG. 2A

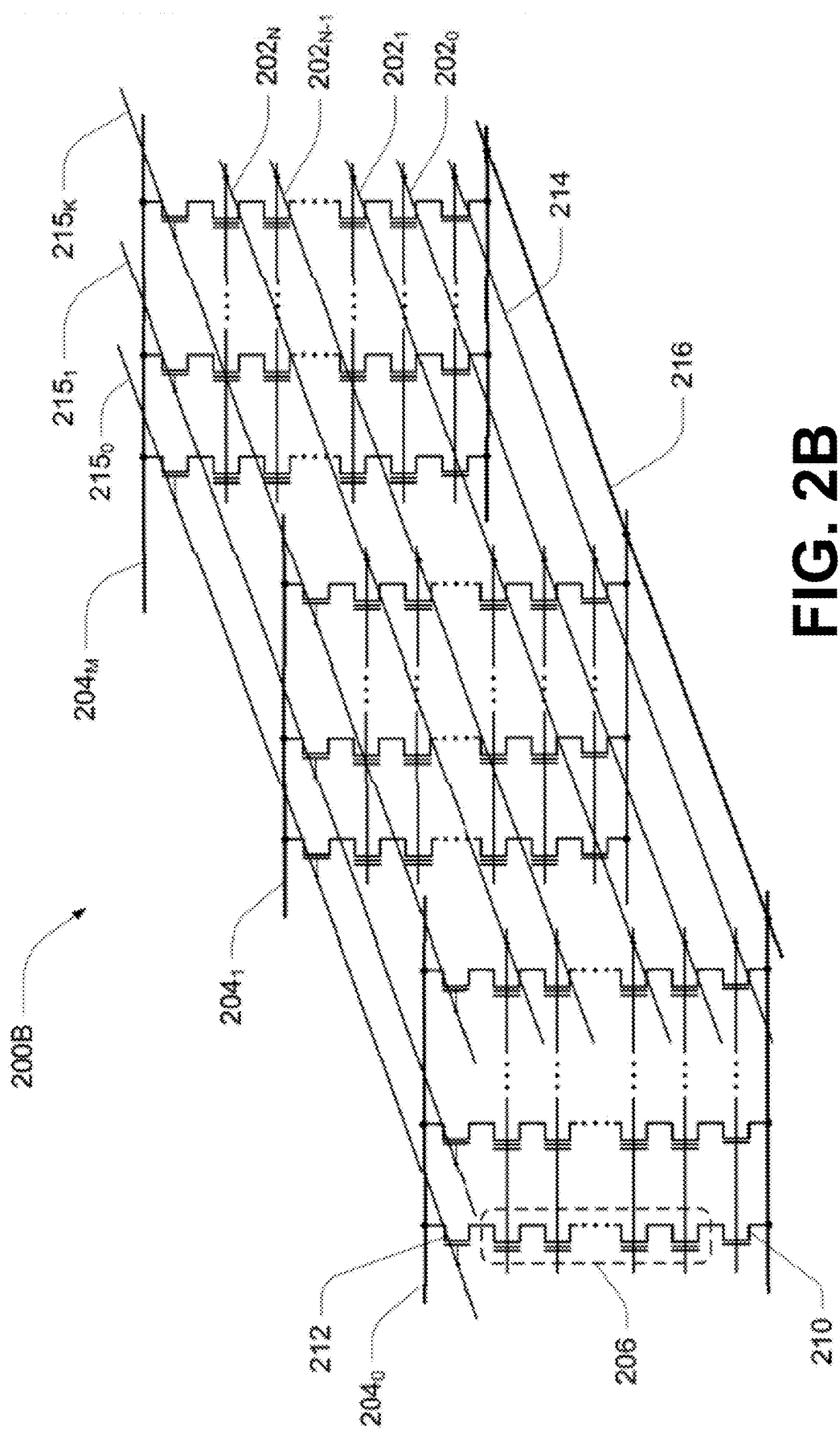


FIG. 2B

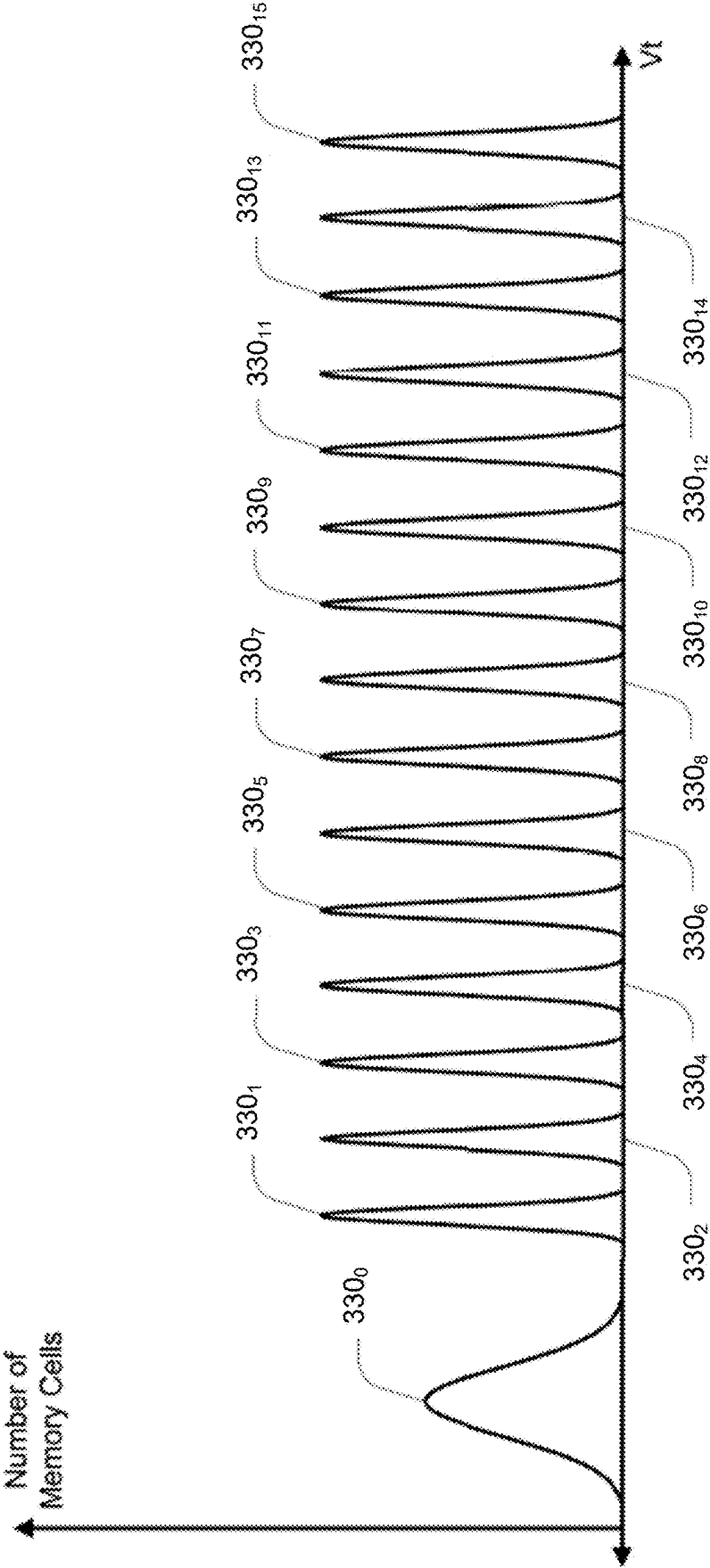


FIG. 3

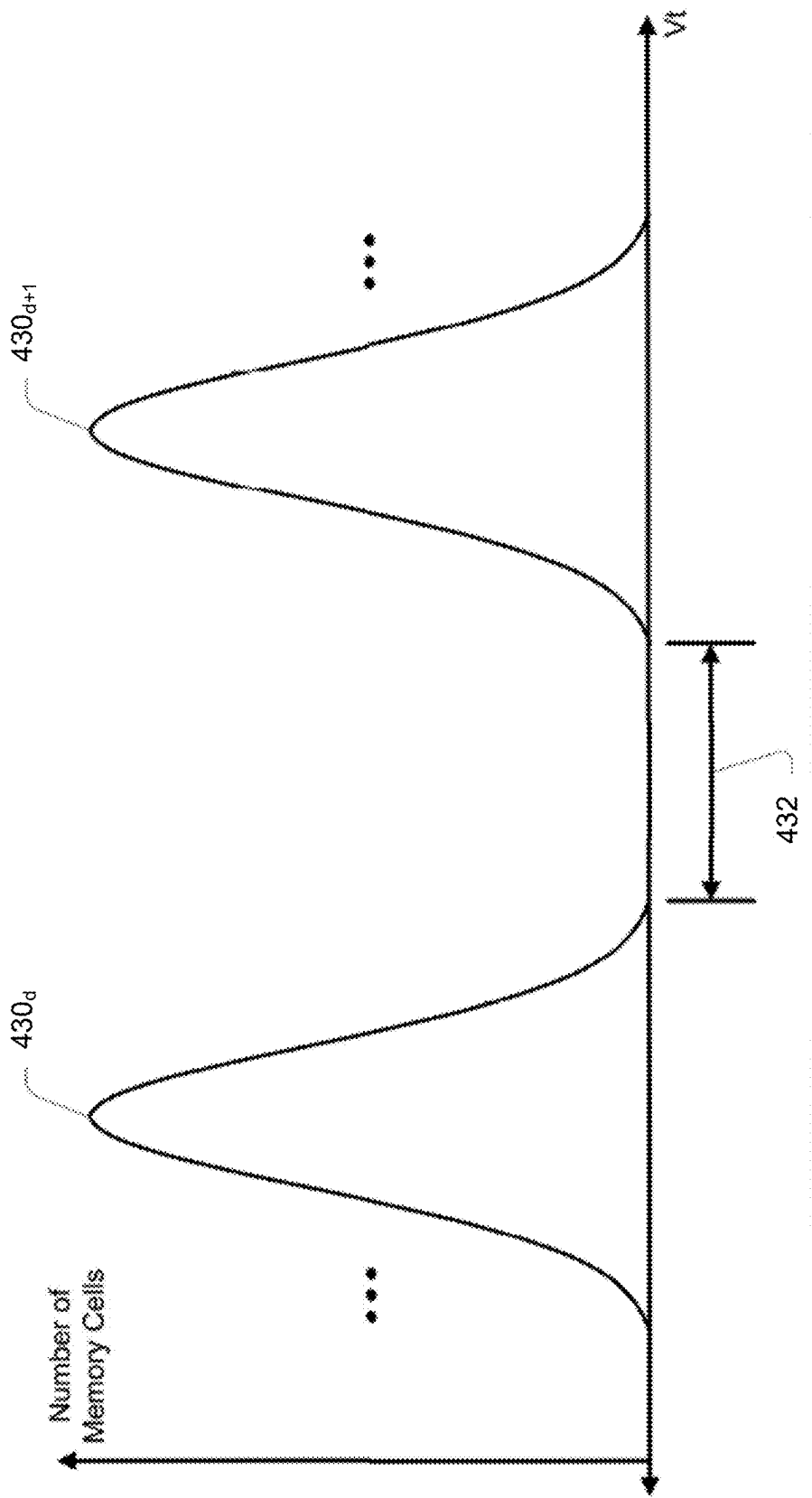
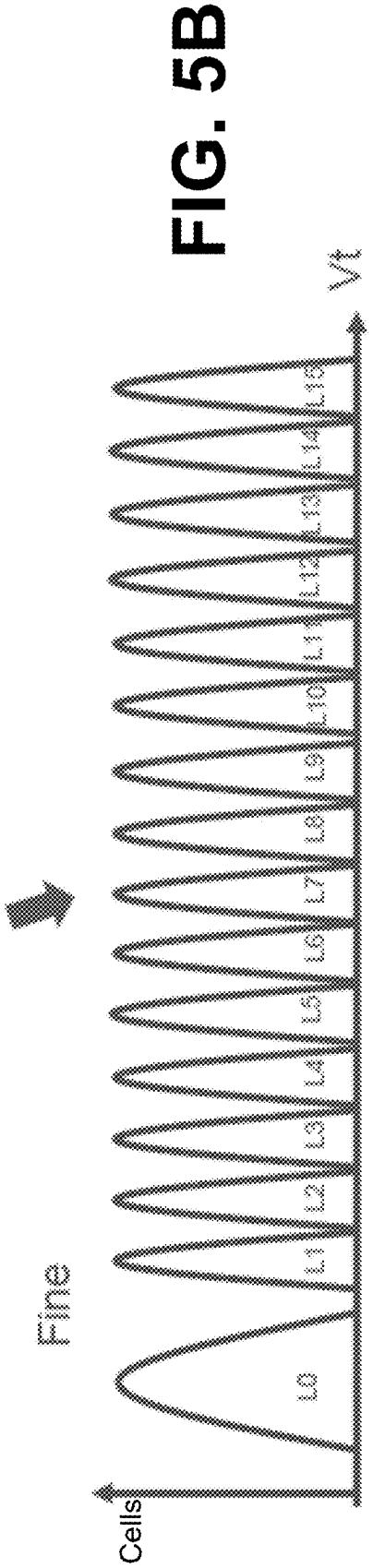
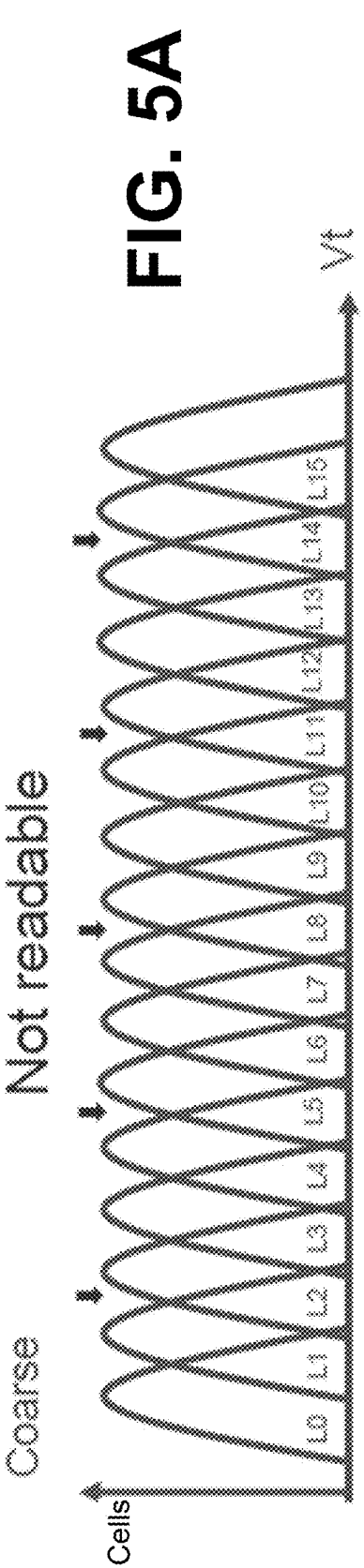


FIG. 4



600

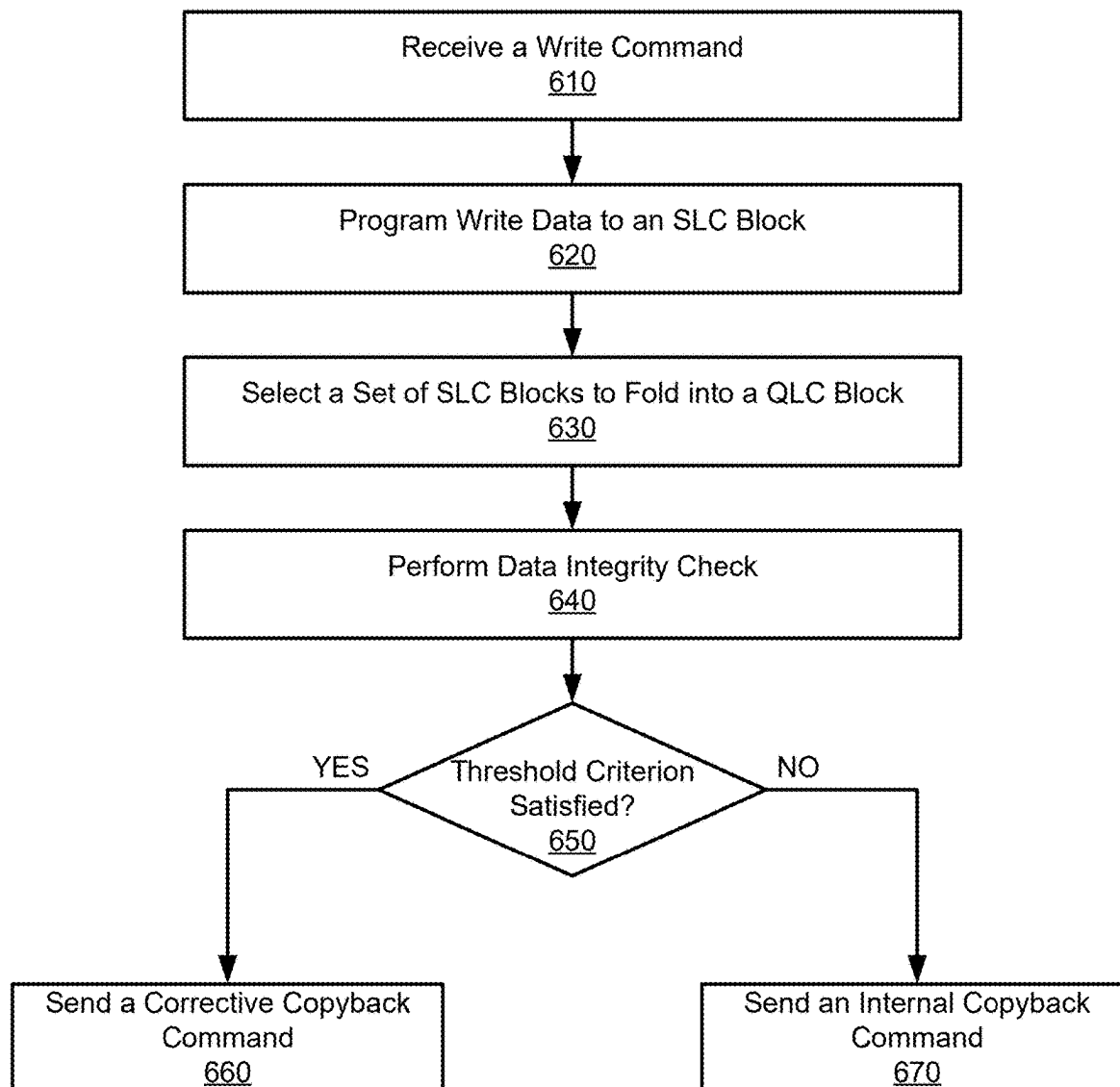
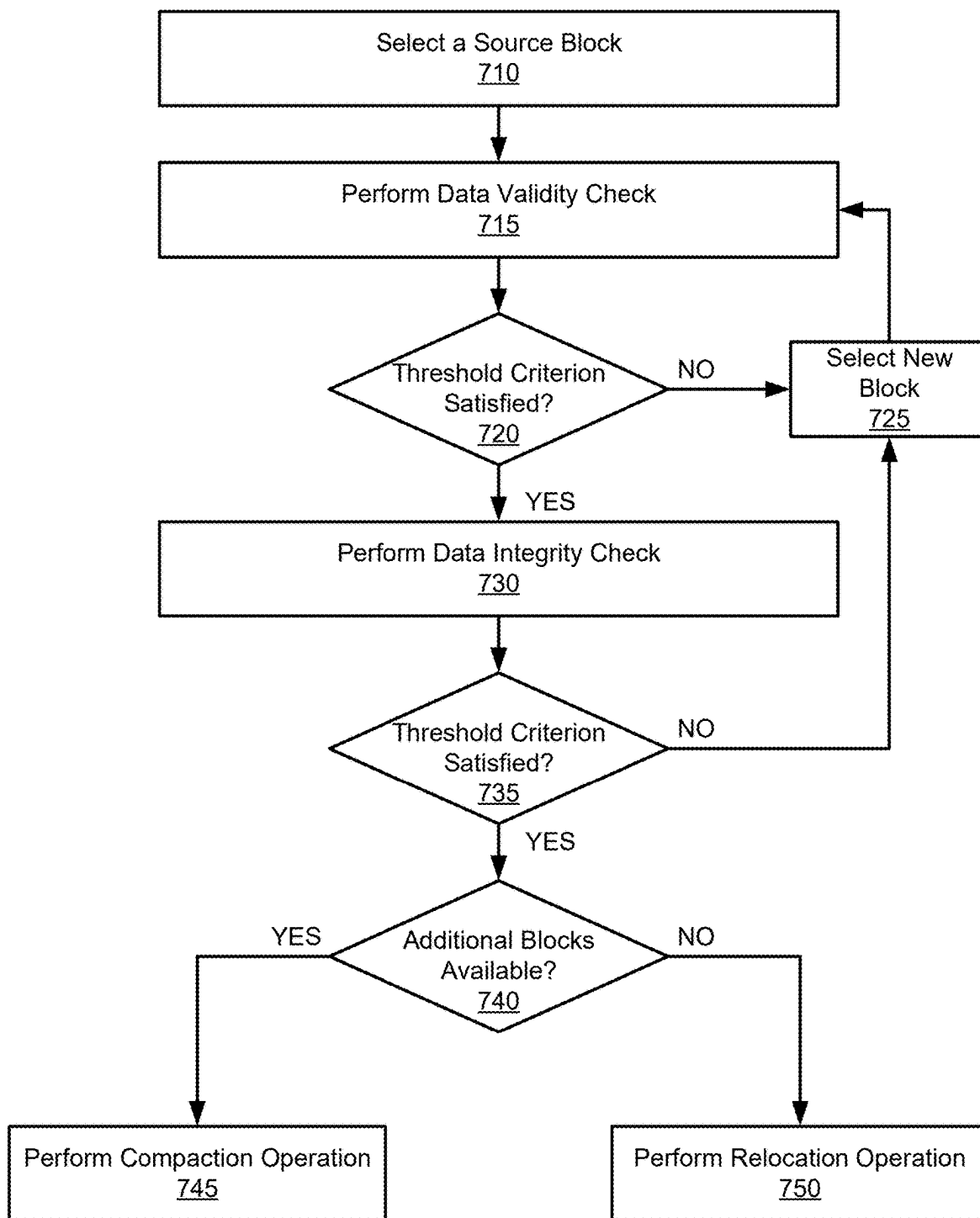


FIG. 6

700**FIG. 7**

800

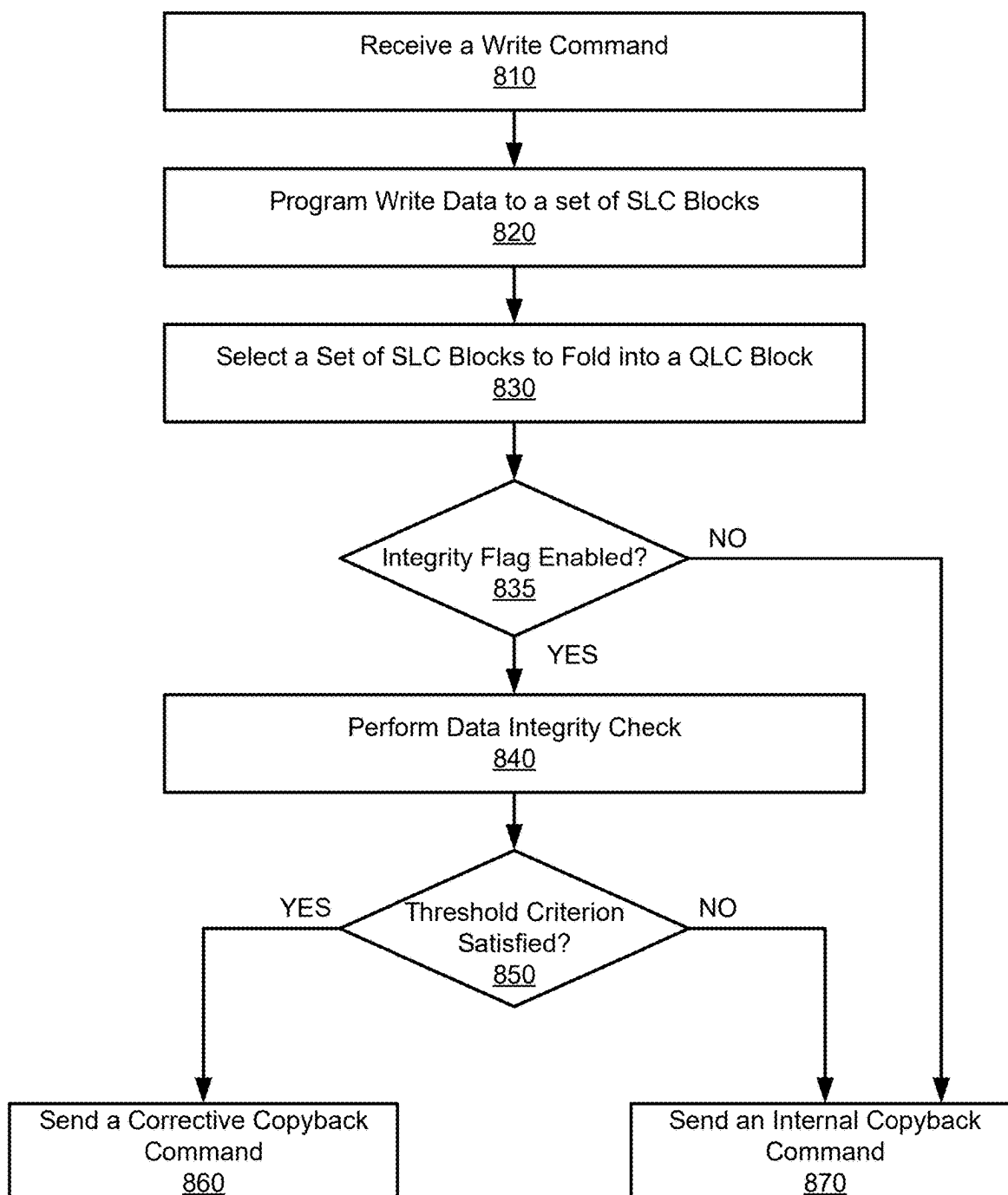


FIG. 8

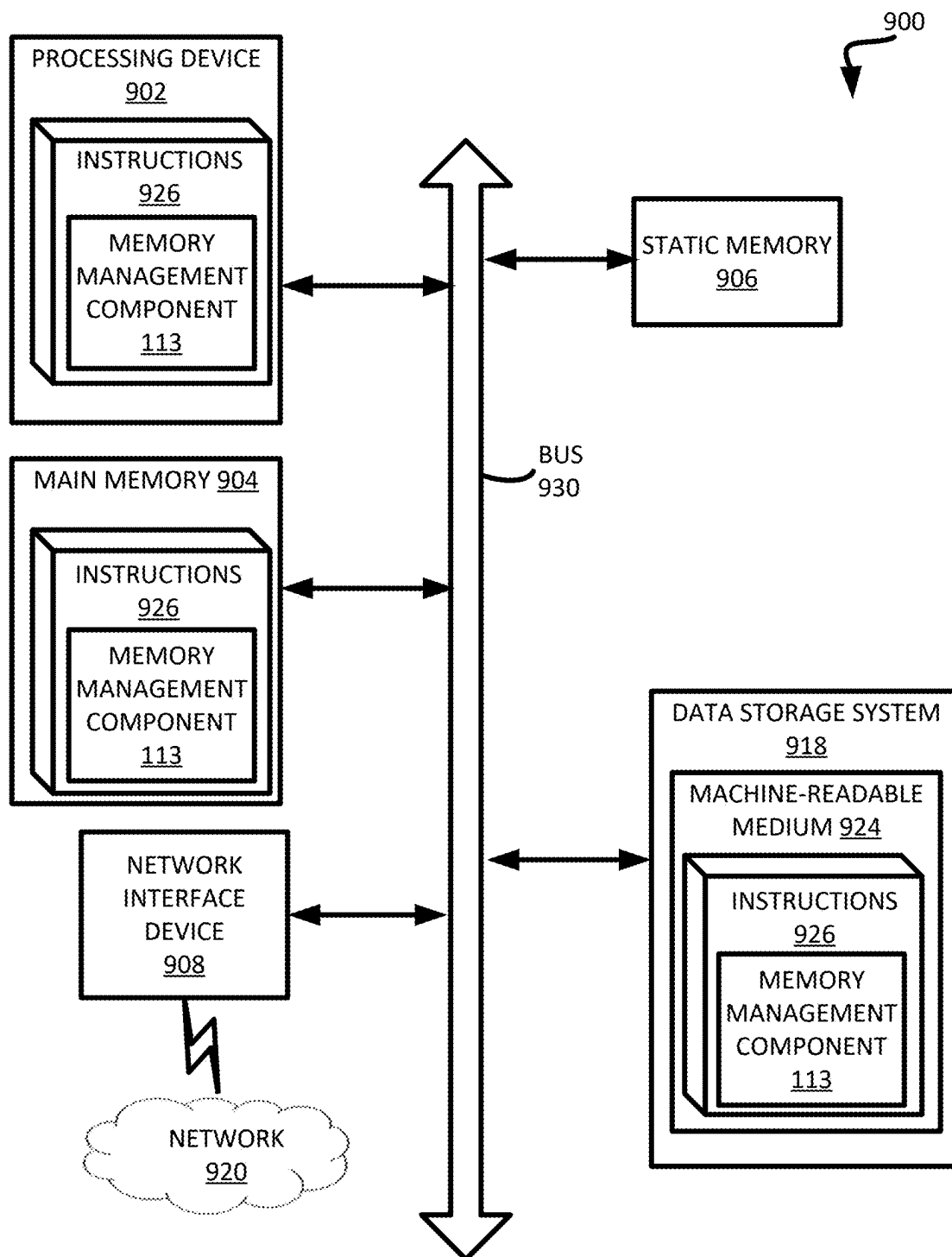


FIG. 9

MEMORY COMPACTION MANAGEMENT IN MEMORY DEVICES

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/432,326, filed Feb. 5, 2024, which is a continuation of U.S. patent application Ser. No. 17/859,468, filed Jul. 7, 2022, now U.S. Pat. No. 11,934,266, issued Mar. 19, 2024, which claims the benefit of U.S. Provisional Application No. 63/347,823, filed Jun. 1, 2022, the entire contents of each is hereby incorporated by reference herein.

TECHNICAL FIELD

[0002] Embodiments of the disclosure relate generally to memory sub-systems, and more specifically, relate to memory compaction management in memory devices.

BACKGROUND

[0003] A memory sub-system can include one or more memory devices that store data. The memory devices can be, for example, non-volatile memory devices and volatile memory devices. In general, a host system can utilize a memory sub-system to store data at the memory devices and to retrieve data from the memory devices.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure.

[0005] FIG. 1A illustrates an example computing system that includes a memory sub-system in accordance with some embodiments of the present disclosure.

[0006] FIG. 1B is a block diagram of a memory device in communication with a memory sub-system controller of a memory sub-system in accordance with some embodiments of the present disclosure.

[0007] FIG. 2A-2B are schematics of portions of an array of memory cells as could be used in a memory of the type described with reference to FIG. 1B in accordance with some embodiments of the present disclosure.

[0008] FIG. 3 is a conceptual depiction of threshold voltage distributions of multiple memory cells of a memory array according to an embodiment.

[0009] FIG. 4 is a conceptual depiction of a threshold voltage distribution of multiple memory cells at one stage following programming for use with various embodiments.

[0010] FIG. 5A is a graph illustrating an example of a set of threshold voltage distributions, after coarse programming, that are not readable according to at least one embodiment.

[0011] FIG. 5B is a graph illustrating the example of the set of threshold voltage distributions, after fine programming, that are readable according to at least one embodiment.

[0012] FIG. 6 is a flow diagram of an example method for performing data compaction in response to a write command, in accordance with some embodiments of the present disclosure.

[0013] FIG. 7 is a flow diagram of an example method for performing compact management during background maintenance operations, in accordance with some embodiments of the present disclosure.

[0014] FIG. 8 is a flow diagram of another example method for performing data compaction in response to a write command, in accordance with some embodiments of the present disclosure.

[0015] FIG. 9 is a block diagram of an example computer system in which embodiments of the present disclosure can operate.

DETAILED DESCRIPTION

[0016] Aspects of the present disclosure are directed to memory compaction management in memory devices. A memory sub-system can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of storage devices and memory modules are described below in conjunction with FIG. 1A. In general, a host system can utilize a memory sub-system that includes one or more components, such as memory devices that store data. The host system can provide data to be stored at the memory sub-system and can request data to be retrieved from the memory sub-system.

[0017] A memory sub-system can include high density non-volatile memory devices where retention of data is desired when no power is supplied to the memory device. One example of non-volatile memory devices is a negative-and (NAND) memory device. Other examples of non-volatile memory devices are described below in conjunction with FIGS. 1A-1B. A non-volatile memory device is a package of one or more dies. Each die can include two or more planes. For some types of non-volatile memory devices (e.g., NAND devices), each plane includes a set of physical blocks. In some implementations, each block can include multiple sub-blocks. Each plane carries a matrix of memory cells formed on a silicon wafer and joined by conductors referred to as wordlines (WLs) and bitlines (BLs), such that a wordline joins multiple memory cells forming a row of the matrix of memory cells, while a bitline joins multiple memory cells forming a column of the matrix of memory cells.

[0018] Depending on the cell type, each memory cell can store one or more bits of binary information, and has various logic states that correlate to the number of bits being stored. The logic states can be represented by binary values, such as "0" and "1," or combinations of such values. A memory cell can be programmed (written to) by applying a certain voltage to the memory cell, which results in an electric charge being held by the memory cell, thus allowing modulation of the voltage distributions produced by the memory cell. A set of memory cells referred to as a memory page can be programmed together in a single operation, e.g., by selecting consecutive bitlines.

[0019] A memory sub-system can use a striping scheme to treat various sets of data as units when performing data operations (e.g., write, read, erase, etc.). A die stripe refers to a collection of planes that are treated as one unit when writing, reading, or erasing data. A controller of a memory device (i.e., a memory sub-system controller, a memory device controller, etc.) can execute the same operation can carry out the same operation, in parallel, at each plane of a die stripe. A block stripe is a collection of blocks, at least one from each plane of a die stripe, that are treated as a unit. The blocks in a block stripe can be associated with the same block identifier (e.g., block number) at each respective

plane. A page stripe is a set of pages having the same page identifier (e.g., the same page number), across a block stripe, and treated as a unit.

[0020] Precisely controlling the amount of the electric charge stored by the memory cell allows establishing multiple logical levels, thus effectively allowing a single memory cell to store multiple bits of information. A read operation can be performed by comparing the measured threshold voltages (V_t) exhibited by the memory cell to one or more reference voltage levels in order to distinguish between two logical levels for single-level cell (SLCs) and between multiple logical levels for multi-level cells. A memory device can include multiple portions, including, e.g., one or more portions where the sub-blocks are configured as SLC memory and one or more portions where the sub-blocks are configured as multi-level cell (MLC) memory that can store three bits of information per cell and/or (triple-level cell) TLC memory that can store three bits of information per cell. The voltage levels of the memory cells in TLC memory form a set of 8 programming (or threshold voltage (V_t)) distributions representing the 8 different combinations of the three bits stored in each memory cell. Depending on how the memory cells are configured, each physical memory page in one of the sub-blocks can include multiple page types. For example, a physical memory page formed from single level cells (SLCs) has a single page type referred to as a lower logical page (LP). Multi-level cell (MLC) physical page types can include LPs and upper logical pages (UPs) and store two bits of information per cell. Further, TLC physical page types can include LPs, UPs, and extra logical pages (XPs) and store three bits of information per cell. Further, quad-level (QLC) physical page types can include LPs, UPs, XPs and top logical pages (TPs) and store four bits of information per cell. For example, a physical memory page formed from memory cells of the QLC memory type have a total of four logical pages, where each logical page stores data distinct from the data stored in the other logical pages associated with that physical memory page, herein referred to as a “page.” MLC memory, TLC memory, QLC memory and PLC memory can be referred to as higher-level cell (HLC) memory.

[0021] A memory device typically experiences random workloads, which can impact the threshold voltage (V_t) distributions, which can be shifted to higher or lower values. Further, the read window budget (RWB), e.g., formed from the read window margins (FIG. 4) between threshold voltage distributions, can be impacted by cell-to-cell interference. If these read window margins are degraded from widening V_t distributions, it can be difficult to read the logical states out of the memory cells, resulting in an increased read bit error rate (RBER). Cell-to-cell interference occurs when a next logical program of memory cells (e.g., coupled with $WLn+1$) causes the threshold voltage (V_t) distribution of victim memory cells (e.g., coupled with WLn) to widen, thus degrading the read window margins between V_t distributions of the WLn victim memory cells. In this scenario, the memory cells associated with $WLn+1$ are physically adjacent to (e.g., above) the memory cells associated with WLn . This cell-to-cell interference imparted to the WLn memory cells is a function of the $WLn+1$ aggressor memory cells incurring a threshold voltage (V_t) swing when being programmed from an erase state to a set of threshold voltage distributions.

[0022] Thus, to minimize cell-to-cell interference, certain memory devices utilize a two-pass, coarse-fine programming algorithm, in which the memory cells associated with WLn first undergo a coarse programming that results in unreadable V_t distributions (FIG. 5A). Then, later, after the aggressor memory cells associated with $WLn+1$ have also been coarse programmed, the memory cells associated with WLn are fine programmed. When this alternating nature of coarse-to-fine programming across wordlines is followed to store a page of data along WLn and another page of data along $WLn+1$, final read window margins of a set of threshold voltage distributions become readable (FIG. 5B). In this way, there is a transition by the $WLn+1$ aggressor memory cells being programmed from an erased V_t distribution to a coarse set of V_t distributions instead of going directly from the erased V_t distribution to the final set of fine V_t distributions, enabling reduction in the cell-to-cell interference as programming progresses due to incurring smaller V_t swings with each programming operation. With a coarse-fine programming algorithm, for example, read window margins for QLC-programmed memory cells can be significantly improved to achieve maximum bits-per-cell for bit-cost scaling. Further, this coarse-fine programming algorithm can help reduce quick charge loss (QCL), which is also referred to as temporal voltage shift, by reducing programmed V_t swings of memory cells.

[0023] The coarse-fine programming algorithm can use SLC memory that is dedicated to buffer QLC data that is being programmed. This buffering of data is required because coarse-programmed V_t distributions have heavily-overlapped states (FIG. 5A), and thus are unreadable while waiting to undergo fine programming into final V_t distributions that are readable (FIG. 5B). Thus, the QLC data is first buffered in SLC blocks of memory so the QLC data is available for the fine programming. For example, according to one coarse-fine programming algorithm of QLC memory, at least 5 coarse QLC pages of data (or approximately 20 SLC pages of data) to be programmed across two adjacent wordlines need to be buffered in SLC blocks.

[0024] Accordingly, to perform the coarse-fine programming, some memory systems use a folding operation in which data is copied from the SLC memory to the QLC memory (or any HLC memory). The folding process allows for compacting data from the SLC memory into the QLC memory, as multiple bits of SLC data can be stored within a single cell of the QLC memory.

[0025] In some memory devices, a folding operation involves data transfers from the memory device to the memory sub-system controller via a data bus (such as an open NAND flash interface or “ONFI” bus), where valid data is read from a source block, subject to error correction (via, for example, ECC) on the memory sub-system controller, and programmed back to a destination block on the memory device. However, this means that a large amount of SLC data is read out and error checked before the controller can proceed with directing the memory device to perform the folding. More so, the data from the SLC memory is read out twice, error checked twice, and programmed twice due to the coarse-fine programming algorithm. This can cause memory devices to incur significant performance penalties in latency, e.g., in the scanning and reading of the SLC data to clear the SLC data to be copied back into HLC memory, and in data bus congestion.

[0026] Aspects of the present disclosure address the above and other deficiencies by causing the memory sub-system controller to perform memory compaction management in a memory device. In particular, the memory sub-system controller can perform operations to optimize the overall performance of the memory device by selecting a particular copyback operation for memory compaction. The copyback operation can be a corrective copyback operation or an internal copyback operation. The corrective copyback operation can include sending data from a source block to a memory sub-system controller to perform error corrections or a refresh of the data stored in the source memory cells before programming the data to a destination block. For example, the memory sub-system controller can perform a scan of the data by reading the data out from the memory device, decoding the source data according to an error correction code (ECC) algorithm, and performing an error check on the source data. The memory sub-system controller can then use one or more error handling mechanisms to correct any detected errors. Once the data is corrected, the memory sub-system controller can send the corrected data to the memory device to be programmed to the destination block. The internal copyback operation can include folding data from the source memory cells to the destination memory cells directly, without intervention by the memory sub-system controller.

[0027] In an illustrative example, in response to a write command, the memory sub-system controller can program the data reflected in the write command to an SLC cache on the memory device. The memory sub-system controller can then select a set of SLC data to program (e.g., via two-pass, coarse-fine programming) into QLC memory. For example, the memory sub-system controller can select four SLC blocks from the SLC cache to program into a QLC block. In other embodiments, any other granularities of memory can be used, such as stripes (e.g., SLC stripes, QLC stripes, etc., where, for example, an SLC stripe and a QLC stripe can be a collection of SLC memory pages and QLC memory pages, respectively, that are associated with the same respective identifier across multiple planes), memory pages (SLC memory pages, QLC memory pages, etc.), etc.

[0028] The memory sub-system controller (or memory device logic) can perform a data integrity check on a portion of the SLC blocks. For example, the data integrity check can include the memory sub-system controller reading data from a set of sampled memory cells in the SLC blocks. The data integrity check can verify that the data stored at memory cells does not include any errors, or that the number of errors are suitably low. In one example, the memory sub-system controller identifies a data integrity metric, such as the bit error count (BEC) or the raw bit error rate (RBER), representing a number of bit errors per unit of time that the data stored at the data block experiences. The memory sub-system controller can then determine whether a value of the data integrity metric (e.g., a BEC value, a RBER value, etc.) satisfies a threshold criterion (e.g., meets or exceeds a threshold value). If the data integrity metric fails to satisfy the threshold criterion (e.g., BEC or RBER value is below a threshold value), the memory sub-system controller can instruct the memory device to perform an internal copyback with respect to the set of SLC blocks. The memory device can then program the data from the set of SLC blocks to a QLC block. This allows the memory device to determine the integrity of the data prior to performing an internal copyback

operation since errors in SLC data can be amplified and thus become uncorrectable once copied into QLC memory. By determining which copyback operation to use, the memory sub-system controller is therefore capable of ensuring the SLC data is sufficiently error free prior to programming the data into QLC memory.

[0029] If the data integrity metric satisfies a threshold criterion (e.g., BEC or RBER value is above a threshold value), indicating a high error rate associated with data stored at the SLC blocks, the memory sub-system controller sends, to the memory device, a corrective copyback command. A corrective copyback operation includes sending, to the memory device, instructions to wait for the memory sub-system controller to perform error correction operations or a refresh of the data stored in the memory cells of the source block before initiating a folding operation of the source data (e.g., the SLC blocks). In some embodiments, the memory sub-system controller performs a scan of the data by reading the data out from the memory device, decoding the source data according to an error correction code (ECC) algorithm, and performing an error check on the source data. The memory sub-system controller can then use one or more error handling mechanisms to correct any detected errors. Once the data is corrected, the memory sub-system controller can send the corrected data to the memory device to be programmed to the destination block (e.g., the QLC block).

[0030] In another illustrative example, the memory sub-system controller can perform operations to optimize the overall performance of the memory device and select a block (source block) on which to perform a reliability management operation. The reliability management operation can perform a compression operation in response to determining that the data on a SLC block is at risk of becoming degraded. In some embodiments, the memory sub-system controller can select a SLC block from the SLC cache. The memory sub-system controller can then perform a data validity check on the SLC block. Data validity can be indicative of a rate of access to the block and/or the age of the data stored on the block. In some embodiments, data validity can be measured using a validity metric based on program/erase cycles (PECs), time after program (TAP), a block read count, etc. For example, the memory sub-system controller can perform a data validity check by obtaining a value indicative of the amount of PECs that occurred with respect to the block, a value indicative of TAP of the block, the read count of the block, etc. In some embodiments, the data validity check can determine a composite value that is a mathematical function (e.g., a weighted sum) of values of two or more validity metrics.

[0031] The memory sub-system controller can then determine whether the data validity value satisfies a threshold criterion. If the data validity metric fails to satisfy (e.g., is lower than) the threshold criterion (e.g., a read count threshold value, a PEC threshold value, and/or TAP threshold value), indicating a relatively low rate of access and/or relatively young data stored on the source block, the memory sub-system controller can select another SLC block in the SLC cache.

[0032] If the data validity metric satisfies (e.g., meets or exceeds) a threshold criterion (e.g., a read count threshold value, a PEC threshold value, and/or TAP threshold value), indicating a relatively high rate of access and/or relatively old data stored on the source block, the memory sub-system

controller can perform a data integrity check on the source block. If the data integrity metric fails to satisfy a threshold criterion (e.g., BEC or RBER value is below a threshold value), indicating a low error rate associated with data stored at the SLC block, the memory sub-system controller moves on to the next SLC block in the SLC cache.

[0033] If the data integrity metric satisfies a threshold criterion (e.g., BEC or RBER value is above a threshold value), indicating a high error rate associated with data stored at the SLC block, the memory sub-system controller determines whether the SLC cache include enough SLC blocks to compact into a QLC block. Responsive to determining that the SLC cache include enough SCL blocks (e.g., four or more SLC blocks), the memory sub-system controller instructs the memory device to compact four SLC blocks into a QLC block. To compact the four SLC blocks, the memory sub-system may perform operation similar to those performed in response to receiving a write command (e.g., determine whether to perform an internal copyback operation or a corrective copyback operation). Responsive to determining that the SLC cache does not include enough SCL blocks (e.g., less than four SLC blocks), the memory sub-system controller can reprogram the SLC block to a different location to refresh or correct the data.

[0034] Advantages of the systems and methods implemented in accordance with some embodiments of the present disclosure include, but are not limited to, a reduction in complexity and improvement in performance of compacting data stored in SLC memory cells to QLC memory cells during programming operations (e.g., coarse-fine programming). This increase in performance includes reduced latency, particularly when the processing device needs take no further action before the internal copyback is performed, and less congestion on a data bus (e.g., ONFI bus) located between the memory sub-system controller and the memory device when avoiding reading out and error checking the SLC data. Other advantages will be apparent to those skilled in the art of folding data within memory devices, which will be discussed hereinafter. Although embodiments are described using wordlines of a NAND flash memory, aspects of the present disclosure can be applied to other types of memory sub-systems.

[0035] FIG. 1A illustrates an example computing system 100 that includes a memory sub-system 110 in accordance with some embodiments of the present disclosure. The memory sub-system 110 can include media, such as one or more volatile memory devices (e.g., memory device 140), one or more non-volatile memory devices (e.g., memory device 130), or a combination of such.

[0036] A memory sub-system 110 can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of a storage device include a solid-state drive (SSD), a flash drive, a universal serial bus (USB) flash drive, an embedded Multi-Media Controller (eMMC) drive, a Universal Flash Storage (UFS) drive, a secure digital (SD) card, and a hard disk drive (HDD). Examples of memory modules include a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), and various types of non-volatile dual in-line memory modules (NVDIMMs).

[0037] The computing system 100 can be a computing device such as a desktop computer, laptop computer, network server, mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), Internet of Things

(IoT) enabled device, embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked commercial device), or such computing device that includes memory and a processing device.

[0038] The computing system 100 can include a host system 120 that is coupled to one or more memory sub-systems 110. In some embodiments, the host system 120 is coupled to different types of memory sub-system 110. FIG. 1A illustrates one example of a host system 120 coupled to one memory sub-system 110. As used herein, “coupled to” or “coupled with” generally refers to a connection between components, which can be an indirect communicative connection or direct communicative connection (e.g., without intervening components), whether wired or wireless, including connections such as electrical, optical, magnetic, etc.

[0039] The host system 120 can include a processor chip-set and a software stack executed by the processor chip-set. The processor chipset can include one or more cores, one or more caches, a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., PCIe controller, SATA controller). The host system 120 uses the memory sub-system 110, for example, to write data to the memory sub-system 110 and read data from the memory sub-system 110.

[0040] The host system 120 can be coupled to the memory sub-system 110 via a physical host interface. Examples of a physical host interface include, but are not limited to, a serial advanced technology attachment (SATA) interface, a peripheral component interconnect express (PCIe) interface, universal serial bus (USB) interface, Fibre Channel, Serial Attached SCSI (SAS), a double data rate (DDR) memory bus, Small Computer System Interface (SCSI), a dual in-line memory module (DIMM) interface (e.g., DIMM socket interface that supports Double Data Rate (DDR)), etc. The physical host interface can be used to transmit data between the host system 120 and the memory sub-system 110. The host system 120 can further utilize an NVM Express (NVMe) interface to access components (e.g., memory devices 130) when the memory sub-system 110 is coupled with the host system 120 by the physical host interface (e.g., PCIe bus). The physical host interface can provide an interface for passing control, address, data, and other signals between the memory sub-system 110 and the host system 120. FIG. 1A illustrates a memory sub-system 110 as an example. In general, the host system 120 can access multiple memory sub-systems via a same communication connection, multiple separate communication connections, and/or a combination of communication connections.

[0041] The memory devices 130, 140 can include any combination of the different types of non-volatile memory devices and/or volatile memory devices. The volatile memory devices (e.g., memory device 140) can be, but are not limited to, random access memory (RAM), such as dynamic random access memory (DRAM) and synchronous dynamic random access memory (SDRAM).

[0042] Some examples of non-volatile memory devices (e.g., memory device 130) include negative-and (NAND) type flash memory and write-in-place memory, such as a three-dimensional cross-point (“3D cross-point”) memory device, which is a cross-point array of non-volatile memory cells. A cross-point array of non-volatile memory can perform bit storage based on a change of bulk resistance, in conjunction with a stackable cross-gridded data access array. Additionally, in contrast to many flash-based memories,

cross-point non-volatile memory can perform a write in-place operation, where a non-volatile memory cell can be programmed without the non-volatile memory cell being previously erased. NAND type flash memory includes, for example, two-dimensional NAND (2D NAND) and three-dimensional NAND (3D NAND).

[0043] Each of the memory devices **130** can include one or more arrays of memory cells. One type of memory cell, for example, single level cells (SLCs) can store one bit per cell. Other types of memory cells, such as multi-level cells (MLCs), triple level cells (TLCs), quad-level cells (QLCs), and penta-level cells (PLCs) can store multiple bits per cell. In some embodiments, each of the memory devices **130** can include one or more arrays of memory cells such as SLCs, MLCs, TLCs, QLCs, PLCs or any combination of such. In some embodiments, a particular memory device can include an SLC portion, and an MLC portion, a TLC portion, a QLC portion, or a PLC portion of memory cells. The memory cells of the memory devices **130** can be grouped as pages that can refer to a logical unit of the memory device used to store data. With some types of memory (e.g., NAND), pages can be grouped to form blocks.

[0044] Although non-volatile memory components such as 3D cross-point array of non-volatile memory cells and NAND type flash memory (e.g. 2D NAND, 3D NAND) are described, the memory device **130** can be based on any other type of non-volatile memory, such as read-only memory (ROM), phase change memory (PCM), self-selecting memory, other chalcogenide based memories, ferroelectric transistor random access memory (FeTRAM), ferroelectric random access memory (FeRAM), magneto random access memory (MRAM), Spin Transfer Torque (STT)-MRAM, conductive bridging RAM (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), negative-or (NOR) flash memory, and electrically erasable programmable read-only memory (EEPROM).

[0045] The memory sub-system controller **115** (or controller **115** for simplicity) can communicate with the memory devices **130** to perform operations such as reading data, writing data, or erasing data at the memory devices **130** and other such operations. The memory sub-system controller **115** can include hardware such as one or more integrated circuits and/or discrete components, a buffer memory, or a combination thereof. The hardware can include a digital circuitry with dedicated (i.e., hard-coded) logic to perform the operations described herein. The memory sub-system controller **115** can be a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc.), or other suitable processor.

[0046] The memory sub-system controller **115** can be a processing device, which includes one or more processors (e.g., processor **117**), configured to execute instructions stored in local memory **119**. In the illustrated example, the local memory **119** of the memory sub-system controller **115** includes an embedded memory configured to store instructions for performing various processes, operations, logic flows, and routines that control operation of the memory sub-system **110**, including handling communications between the memory sub-system **110** and the host system **120**.

[0047] In the illustrated example, the local memory **119** of the memory sub-system controller **115** includes an embedded memory configured to store instructions for performing

various processes, operations, logic flows, and routines that control operation of the memory sub-system **110**, including handling communications between the memory sub-system **110** and the host system **120**.

[0048] In some embodiments, the local memory **119** can include memory registers storing memory pointers, fetched data, etc. The local memory **119** can also include read-only memory (ROM) for storing micro-code. While the example memory sub-system **110** in FIG. 1A has been illustrated as including the memory sub-system controller **115**, in another embodiment of the present disclosure, a memory sub-system **110** does not include a memory sub-system controller **115**, and can instead rely upon external control (e.g., provided by an external host, or by a processor or controller separate from the memory sub-system).

[0049] In general, the memory sub-system controller **115** can receive commands or operations from the host system **120** and can convert the commands or operations into instructions or appropriate commands to achieve the desired access to the memory devices **130**. The memory sub-system controller **115** can be responsible for other operations such as wear leveling operations, garbage collection operations, error detection and error-correcting code (ECC) operations, encryption operations, caching operations, and address translations between a logical address (e.g., logical block address (LBA), namespace) and a physical address (e.g., physical MU address, physical block address) that are associated with the memory devices **130**. The memory sub-system controller **115** can further include host interface circuitry to communicate with the host system **120** via the physical host interface. The host interface circuitry can convert the commands received from the host system into command instructions to access the memory devices **130** as well as convert responses associated with the memory devices **130** into information for the host system **120**.

[0050] The memory sub-system **110** can also include additional circuitry or components that are not illustrated. In some embodiments, the memory sub-system **110** can include a cache or buffer (e.g., DRAM) and address circuitry (e.g., a row decoder and a column decoder) that can receive an address from the memory sub-system controller **115** and decode the address to access the memory devices **130**.

[0051] In some embodiments, the memory devices **130** include local media controllers **135** that operate in conjunction with memory sub-system controller **115** to execute operations on one or more memory cells of the memory devices **130**. An external controller (e.g., memory sub-system controller **115**) can externally manage the memory device **130** (e.g., perform media management operations on the memory device **130**). In some embodiments, memory sub-system **110** is a managed memory device, which includes a raw memory device **130** having control logic (e.g., local controller **132**) on the die and a controller (e.g., memory sub-system controller **115**) for media management within the same memory device package. An example of a managed memory device is a managed NAND (MNAND) device.

[0052] In some embodiments, the controller **115** (e.g., processing device) includes an error-correcting code (ECC) encoder/decoder **111**. The ECC encoder/decoder **111** can perform ECC encoding for data written to the memory devices **130** and ECC decoding for data read from the memory devices **130**, respectively. The ECC decoding can be performed to decode an ECC codeword to correct errors

in the raw read data, and in many cases also to report the number of bit errors in the raw read data. The memory sub-system controller **115** can further include a processor **117** (processing device) configured to execute instructions stored in local memory **119** for performing the operations described herein. The local memory **119** can also buffer data being used by the executed instructions. Example ECC processes can include parity checks, Hamming codes, checksums, cyclic redundancy checks (CRCs), cryptographic hashes, block codes, convolutional codes, turbo codes, low-density parity check (LDPC), etc.

[0053] In at least some embodiments, the controller **115** further includes a memory management component **113** that can handle interactions of the controller **115** with the memory devices of the memory sub-system **110**, such as with the memory device **130**. For example, the memory interface component **113** can determine whether to perform an internal copyback operation or a corrective copyback operation with respect to source data stored on a block. An internal copyback operation includes sending an internal copyback command to memory device **130**, **140** with instructions to perform a copyback of data from source memory cells to destination memory cells without further intervention by the controller **115**. A corrective copyback operation includes sending a corrective copyback command to memory device **130**, **140** with instructions to wait for the controller **115** to perform error correction or a refresh of the data stored in the source memory cells before performing a copyback operation. In some embodiments, the memory sub-system controller **115** includes at least a portion of the memory management component **113**. In some embodiments, the memory management component **113** is part of the host system **120**, an application, or an operating system. In other embodiments, local media controller **135** includes at least a portion of memory management component **113** and is configured to perform the functionality described herein. The memory management component **113** can communicate directly with the memory devices **130** and **140** via a synchronous interface. Furthermore, transfers of data between the memory device **130** and the memory device **140** can be done within the memory sub-system **110** without accessing the host system **120**.

[0054] In various embodiments, the memory device **130** further includes one or more page buffers **152**, which can provide the circuitry used to program data to the memory cells of the memory device **130** and to read the data out of the memory cells. The local media controller **135** can further include a program manager **136**, which is implemented using firmware, hardware, or a combination of firmware and hardware. In one embodiment, the program manager **136** receives an internal copyback command and/or a corrective copyback command from the memory management component **113**. The program manager **136** can execute the internal copyback command by performing a copyback of data from source memory cells to destination memory cells without further intervention by the controller **115**, or execute the corrective copyback command by waiting for the controller **115** to perform error correction or a refresh of the data stored in the source memory cells before performing a copyback operation. In some embodiments, the program manager **136** is part of the host system **120**, an application, or an operating system. In some embodiments, control logic of the local media controller **135** and/or program manager **136** can implement or direct the consecutive coarse and fine pro-

gramming operations on a set of memory cells and other related operations described herein. In some embodiments, this control logic is integrated in whole or in part within the memory sub-system controller **115** and/or the host system **120**. Further details with regards to the operations of program manager **136** are described below.

[0055] FIG. 1B is a simplified block diagram of a first apparatus, in the form of a memory device **130**, in communication with a second apparatus, in the form of a memory sub-system controller **115** of a memory sub-system (e.g., the memory sub-system **110** of FIG. 1A), according to an embodiment. Some examples of electronic systems include personal computers, personal digital assistants (PDAs), digital cameras, digital media players, digital recorders, games, appliances, vehicles, wireless devices, mobile telephones and the like. The memory sub-system controller **115** (e.g., a controller external to the memory device **130**), can be a memory controller or other external host device.

[0056] The memory device **130** includes an array of memory cells **104** logically arranged in rows and columns. Memory cells of a logical row are typically connected to the same access line (e.g., a wordline) while memory cells of a logical column are typically selectively connected to the same data line (e.g., a bitline). A single access line can be associated with more than one logical row of memory cells and a single data line can be associated with more than one logical column. Memory cells (not shown in FIG. 1B) of at least a portion of the array of memory cells **104** are capable of being programmed to one of at least two target data states.

[0057] Row decode circuitry **108** and column decode circuitry **109** are provided to decode address signals. Address signals are received and decoded to access the array of memory cells **104**. The memory device **130** also includes input/output (I/O) control circuitry **112** to manage input of commands, addresses and data to the memory device **130** as well as output of data and status information from the memory device **130**. An address register **114** is in communication with the I/O control circuitry **112** and row decode circuitry **108** and column decode circuitry **109** to latch the address signals prior to decoding. A command register **124** is in communication with the I/O control circuitry **112** and local media controller **135** to latch incoming commands.

[0058] A controller (e.g., the local media controller **135** internal to the memory device **130**) controls access to the array of memory cells **104** in response to the commands and generates status information for the external memory sub-system controller **115**, i.e., the local media controller **135** is configured to perform access operations (e.g., read operations, programming operations and/or erase operations) on the array of memory cells **104**. The local media controller **135** is in communication with row decode circuitry **108** and column decode circuitry **109** to control the row decode circuitry **108** and column decode circuitry **109** in response to the addresses. The local media controller **135** can also include the program manager **136**, as was discussed.

[0059] The local media controller **135** is also in communication with a cache register **118** and a data register **121**. The cache register **118** latches data, either incoming or outgoing, as directed by the local media controller **135** to temporarily store data while the array of memory cells **104** is busy writing or reading, respectively, other data. During a program operation (e.g., write operation), data can be passed from the cache register **118** to the data register **121** for transfer to the array of memory cells **104**; then new data

can be latched in the cache register **118** from the I/O control circuitry **112**. During a read operation, data can be passed from the cache register **118** to the I/O control circuitry **112** for output to the memory sub-system controller **115**; then new data can be passed from the data register **121** to the cache register **118**. The cache register **118** and/or the data register **121** can form (e.g., can form at least a portion of) a page buffer of the one or more page buffers **152** of the memory device **130**. Each page buffer can further include sensing devices such as a sense amplifier, to sense a data state of a memory cell of the array of memory cells **104**, e.g., by sensing a state of a data line connected to that memory cell. A status register **122** can be in communication with I/O control circuitry **112** and the local memory controller **135** to latch the status information for output to the memory sub-system controller **115**.

[0060] The memory device **130** receives control signals at the memory sub-system controller **115** from the local media controller **135** over a control link **132**. For example, the control signals can include a chip enable signal CE #, a command latch enable signal CLE, an address latch enable signal ALE, a write enable signal WE #, a read enable signal RE #, and a write protect signal WP #. Additional or alternative control signals (not shown) can be further received over control link **132** depending upon the nature of the memory device **130**. In one embodiment, memory device **130** receives command signals (which represent commands), address signals (which represent addresses), and data signals (which represent data) from the memory sub-system controller **115** over a multiplexed input/output (I/O) bus **134** and outputs data to the memory sub-system controller **115** over I/O bus **134**.

[0061] For example, the commands can be received over input/output (I/O) pins [7:0] of I/O bus **134** at I/O control circuitry **112** and can then be written into a command register **124**. The addresses can be received over input/output (I/O) pins [7:0] of I/O bus **134** at I/O control circuitry **112** and can then be written into address register **114**. The data can be received over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device at I/O control circuitry **112** and then can be written into cache register **118**. The data can be subsequently written into data register **121** for programming the array of memory cells **104**.

[0062] In an embodiment, cache register **118** can be omitted, and the data can be written directly into data register **121**. Data can also be output over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device. Although reference can be made to I/O pins, they can include any conductive node providing for electrical connection to the memory device **130** by an external device (e.g., the memory sub-system controller **115**), such as conductive pads or conductive bumps as are commonly used.

[0063] It will be appreciated by those skilled in the art that additional circuitry and signals can be provided, and that the memory device **130** of FIG. 1B has been simplified. It should be recognized that the functionality of the various block components described with reference to FIG. 1B may not necessarily be segregated to distinct components or component portions of an integrated circuit device. For example, a single component or component portion of an integrated circuit device could be adapted to perform the functionality of more than one block component of FIG. 1B.

Alternatively, one or more components or component portions of an integrated circuit device could be combined to perform the functionality of a single block component of FIG. 1B. Additionally, while specific I/O pins are described in accordance with popular conventions for receipt and output of the various signals, it is noted that other combinations or numbers of I/O pins (or other I/O node structures) can be used in the various embodiments.

[0064] FIG. 2A-2B are schematics of portions of an array of memory cells **200A**, such as a NAND memory array, as could be used in a memory of the type described with reference to FIG. 1B according to an embodiment, e.g., as a portion of the array of memory cells **104**. Memory array **200A** includes access lines, such as wordlines **202₀** to **202_N**, and data lines, such as bitlines **204₀** to **204_M**. The wordlines **202** can be connected to global access lines (e.g., global wordlines), not shown in FIG. 2A, in a many-to-one relationship. For some embodiments, memory array **200A** can be formed over a semiconductor that, for example, can be conductively doped to have a conductivity type, such as a p-type conductivity, e.g., to form a p-well, or an n-type conductivity, e.g., to form an n well.

[0065] Memory array **200A** can be arranged in rows (each corresponding to a wordline **202**) and columns (each corresponding to a bitline **204**). Each column can include a string of series-connected memory cells (e.g., non-volatile memory cells), such as one of NAND strings **206₀** to **206_M**. Each NAND string **206** can be connected (e.g., selectively connected) to a common source (SRC) **216** and can include memory cells **208₀** to **208_N**. Each bitline **204** and NAND string **206** can be associated with a sub-block of a set of sub-blocks of the memory array **200A**. The memory cells **208** can represent non-volatile memory cells for storage of data. The memory cells **208** of each NAND string **206** can be connected in series between a select gate **210** (e.g., a field-effect transistor), such as one of the select gates **210₀** to **210_M** (e.g., that can be source select transistors, commonly referred to as select gate source), and a select gate **212** (e.g., a field-effect transistor), such as one of the select gates **212₀** to **212_M** (e.g., that can be drain select transistors, commonly referred to as select gate drain). Select gates **210₀** to **210_M** can be commonly connected to a select line **214**, such as a source select line (SGS), and select gates **212₀** to **212_M** can be commonly connected to a select line **215**, such as a drain select line (SGD). Although depicted as traditional field-effect transistors, the select gates **210** and **212** can utilize a structure similar to (e.g., the same as) the memory cells **208**. The select gates **210** and **212** can represent a number of select gates connected in series, with each select gate in series configured to receive a same or independent control signal.

[0066] A source of each select gate **210** can be connected to common source **216**. The drain of each select gate **210** can be connected to a memory cell **208₀** of the corresponding NAND string **206**. For example, the drain of select gate **210₀** can be connected to memory cell **208₀** of the corresponding NAND string **206₀**. Therefore, each select gate **210** can be configured to selectively connect a corresponding NAND string **206** to the common source **216**. A control gate of each select gate **210** can be connected to the select line **214**.

[0067] The drain of each select gate **212** can be connected to the bitline **204** for the corresponding NAND string **206**. For example, the drain of select gate **212₀** can be connected to the bitline **204₀** for the corresponding NAND string **206₀**.

The source of each select gate **212** can be connected to a memory cell **208_N** of the corresponding NAND string **206**. For example, the source of select gate **212₀** can be connected to memory cell **208_N** of the corresponding NAND string **206₀**. Therefore, each select gate **212** can be configured to selectively connect a corresponding NAND string **206** to the corresponding bitline **204**. A control gate of each select gate **212** can be connected to select line **215**.

[0068] The memory array **200A** in FIG. 2A can be a quasi-two-dimensional memory array and can have a generally planar structure, e.g., where the common source **216**, NAND strings **206** and bitlines **204** extend in substantially parallel planes. Alternatively, the memory array **200A** in FIG. 2A can be a three-dimensional memory array, e.g., where NAND strings **206** can extend substantially perpendicular to a plane containing the common source **216** and to a plane containing the bitlines **204** that can be substantially parallel to the plane containing the common source **216**.

[0069] Typical construction of memory cells **208** includes a data-storage structure **234** (e.g., a floating gate, charge trap, and the like) that can determine a data state of the memory cell (e.g., through changes in threshold voltage), and a control gate **236**, as shown in FIG. 2A. The data-storage structure **234** can include both conductive and dielectric structures while the control gate **236** is generally formed of one or more conductive materials. In some cases, memory cells **208** can further have a defined source/drain (e.g., source) **230** and a defined source/drain (e.g., drain) **232**. The memory cells **208** have their control gates **236** connected to (and in some cases form) a wordline **202**.

[0070] A column of the memory cells **208** can be a NAND string **206** or a number of NAND strings **206** selectively connected to a given bitline **204**. A row of the memory cells **208** can be memory cells **208** commonly connected to a given wordline **202**. A row of memory cells **208** can, but need not, include all the memory cells **208** commonly connected to a given wordline **202**. Rows of the memory cells **208** can often be divided into one or more groups of physical pages of memory cells **208**, and physical pages of the memory cells **208** often include every other memory cell **208** commonly connected to a given wordline **202**. For example, the memory cells **208** commonly connected to wordline **202_N** and selectively connected to even bitlines **204** (e.g., bitlines **204₀**, **204₂**, **204₄**, etc.) can be one physical page of the memory cells **208** (e.g., even memory cells) while memory cells **208** commonly connected to wordline **202_N** and selectively connected to odd bitlines **204** (e.g., bitlines **204₁**, **204₃**, **204₅**, etc.) can be another physical page of the memory cells **208** (e.g., odd memory cells).

[0071] Although bitlines **204₃**, **204₅** are not explicitly depicted in FIG. 2A, it is apparent from the figure that the bitlines **204** of the array of memory cells **200A** can be numbered consecutively from bitline **204₀** to bitline **204_M**. Other groupings of the memory cells **208** commonly connected to a given wordline **202** can also define a physical page of memory cells **208**. For certain memory devices, all memory cells commonly connected to a given wordline can be deemed a physical page of memory cells. The portion of a physical page of memory cells (which, in some embodiments, could still be the entire row) that is read during a single read operation or programmed during a single programming operation (e.g., an upper or lower page of memory cells) can be deemed a logical page of memory cells. A block of memory cells can include those memory cells that are configured to be erased together, such as all memory cells connected to wordlines **202₀**-**202_N** (e.g., all

NAND strings **206** sharing common wordlines **202**). Unless expressly distinguished, a reference to a page of memory cells herein refers to the memory cells of a logical page of memory cells. Although the example of FIG. 2A is discussed in conjunction with NAND flash, the embodiments and concepts described herein are not limited to a particular array architecture or structure, and can include other structures (e.g., SONOS, phase change, ferroelectric, etc.) and other architectures (e.g., AND arrays, NOR arrays, etc.).

[0072] FIG. 2B is another schematic of a portion of an array of memory cells **200B** as could be used in a memory of the type described with reference to FIG. 1B, e.g., as a portion of the array of memory cells **104**. Like numbered elements in FIG. 2B correspond to the description as provided with respect to FIG. 2A. FIG. 2B provides additional detail of one example of a three-dimensional NAND memory array **200B** can incorporate vertical structures which can include semiconductor pillars where a portion of a pillar can act as a channel region of the memory cells of NAND strings **206**. The NAND strings **206** can be each selectively connected to a bitline **204₀**-**204_M** by a select transistor **212** (e.g., that can be drain select transistors, commonly referred to as select gate drain) and to a common source **216** by a select transistor **210** (e.g., that can be source select transistors, commonly referred to as select gate source). Multiple NAND strings **206** can be selectively connected to the same bitline **204**. Subsets of NAND strings **206** can be connected to their respective bitlines **204** by biasing the select lines **215₀**-**215_K** to selectively activate particular select transistors **212** each between a NAND string **206** and a bitline **204**. The select transistors **210** can be activated by biasing the select line **214**. Each wordline **202** can be connected to multiple rows of memory cells of the memory array **200B**. Rows of memory cells that are commonly connected to each other by a particular wordline **202** can collectively be referred to as tiers.

[0073] FIG. 3 is a conceptual depiction of threshold voltage ranges of multiple memory cells. FIG. 3 illustrates an example of threshold voltage ranges and their distributions for a population of a sixteen-level memory cells, e.g., QLC memory cells. For example, such a memory cell can be programmed to a threshold voltage (V_t) that falls within one of sixteen different threshold voltage ranges **330₀**-**330₁₅**, each being used to represent a data state corresponding to a bit pattern of four bits. The threshold voltage range **330₀** typically has a greater width than the remaining threshold voltage ranges **330₁**-**330₁₅** as memory cells are generally all placed in the data state corresponding to the threshold voltage range **330₀**, then subsets of those memory cells are subsequently programmed to have threshold voltages in one of the threshold voltage ranges **330₁**-**330₁₅**. As programming operations are generally more incrementally controlled than erase operations, these threshold voltage ranges **330₁**-**330₁₅** can tend to have tighter distributions.

TABLE 1

Data State	Logical Data Value
L0	1111
L1	0111
L2	0011
L3	1011

TABLE 1-continued

Data State	Logical Data Value
L4	1001
L5	0001
L6	0101
L7	1101
L8	1100
L9	0100
L10	0000
L11	1000
L12	1010
L13	0010
L14	0110
L15	1110

[0074] The threshold voltage ranges 330_0 , 330_1 , 330_2 , 330_3 , 330_4 , 330_5 , 330_6 , 330_7 , 330_8 , 330_9 , 330_{10} , 330_{11} , 330_{12} , 330_{13} , 330_{14} , and 330_{15} can each represent a respective data state, e.g., L0, L1, L2, L3, L4, L5, L6, L7, L8, L9, L10, L11, L12, L13, L14 and L15, respectively. As an example, if the threshold voltage of a memory cell is within the first of the sixteen threshold voltage ranges 330_0 , the memory cell in this case can be storing a data state L0 having a data value of logical '1111' and is typically referred to as the erased state of the memory cell. If the threshold voltage is within the second of the sixteen threshold voltage ranges 330_1 , the memory cell in this case can be storing a data state L1 having a data value of logical '0111'. If the threshold voltage is within the third of the sixteen threshold voltage ranges 330_2 , the memory cell in this case can be storing a data state L2 having a data value of logical '0011,' and so on. Table 1 provides one possible correspondence between the data states and their corresponding logical data values. Other assignments of data states to logical data values are known or can be envisioned. Memory cells remaining in the lowest data state (e.g., the erased state or L0 data state), as used herein, will be deemed to be programmed to the lowest data state.

[0075] FIG. 4 is a conceptual depiction of a threshold voltage distribution of multiple memory cells following a programming operation. The threshold voltage distributions 430_d - 430_{d+1} of FIG. 4 can represent some portion of the distributions for threshold voltage ranges 330_0 - 330_{15} of FIG. 3 at the completion of a programming operation for memory cells. With reference to FIG. 4, adjacent threshold voltage distributions 430 are typically separated by some read window margin 432 (e.g., dead space) at the completion of programming. Applying a sense voltage (e.g., read level voltage) within the read window margin 432 to the control gates of the multiple memory cells can be used to distinguish between the memory cells of the threshold voltage distribution 430_d (and any lower threshold voltage distribution) and the memory cells of the threshold voltage distribution 430_{d+1} (and any higher threshold voltage distribution).

[0076] Due to the phenomenon known as slow charge loss (SCL), the threshold voltage of a memory cell changes in time as the electric charge of the cell is degrading, which is referred to as "temporal voltage shift" (since the degrading electric charge causes the voltage distributions to shift along the voltage axis towards lower voltage levels). The threshold voltage is changing rapidly at first (immediately after the memory cell was programmed, referred to as quick charge loss (QCL)), and then slows down in an approximately logarithmic linear fashion with respect to the time elapsed

since the cell programming event. This temporal voltage shift, if left unadjusted, reduces the read window margin 432 between the threshold voltage distributions 430_d - 430_{d+1} over time, and can cause these threshold voltage distributions to overlap, making it more difficult to distinguish between adjacent threshold voltage distributions.

[0077] FIG. 5A is a graph illustrating an example of a set of threshold voltage distributions, after coarse programming, that are not readable according to at least one embodiment. Coarse programming can be compared to initial pass programming in which the Vt distributions are highly overlapped when coarse programming many Vt distributions, e.g. as is the case in programming QLC memory. Due to this overlapping of the Vt distributions, the coarse-programmed sets of threshold Vt distributions may also be referred to herein as intermediate Vt distributions. This overlapping occurs due to less precise programming in which each Vt distribution widely covers a range of threshold voltage that coarsely approximates a more accurate (finer) threshold voltage range that is intended for each respective Vt distribution.

[0078] FIG. 5B is a graph illustrating the example of the set of threshold voltage distributions (FIG. 5B), after fine programming, that are readable according to at least one embodiment. When fine programming is completed, e.g., to a final set of Vt distributions, each Vt distribution is more finely defined over a focused threshold voltage range intended for each respective logical state. When this occurs, the read window margins between respective Vt distributions are widened such that individual logical states across different memory cells of a set of memory cells can be distinguished when read.

[0079] While the overlap between coarsely-programmed Vt distributions is more pronounced in QLC programming, the disclosed consecutive coarse and fine programming operations can be performed on differing types of memory where coarse programming can initially be carried out in a fewer number of intermediate Vt distributions followed by consecutive coarse and fine programming of a larger number of final Vt distributions, as will be discussed. Also, as was mentioned, the touch-up effect of performing fine programming on the set of memory cells after a coarse programming of the same number of Vt distributions leads to improved read window margins (or RWB) and QCL.

[0080] FIG. 6 is a flow diagram of an example method 600 for performing data compaction in response to a write command, in accordance with some embodiments of the present disclosure. The method 600 can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method 600 is performed by the media management component 113 of FIG. 1A. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

[0081] The operations of method 600 can relate to performing an internal copyback or a selective copyback of SLC data to QLC data. However, it is noted that the operations of method 600 can be performed between any types of SLC or HLC data (e.g., SLC to HLC, SLC to SLC, HLC to HLC, or HLC to SLC). Method 600 will be discussed in relation to the copyback operations being performed from a set of SLC source blocks to a QLC destination block. However, it should be understood that method 600 can be performed using any sized memory granularity, such as, but not limited to, a set of SLC source blocks and a HLC destination block, a set of SLC source memory pages and a HLC destination memory page, etc.

[0082] At operation 610, the processing logic receives a write command. The write command can be initiated by a host (e.g., host 120) or by a memory sub-system controller (e.g., memory sub-system controller 115). In some embodiments, responsive to receiving the write command, the processing logic can identify a source block(s) in the SLC portion of the memory device (e.g., an SLC cache), and a destination block(s) in a QLC portion of the memory device. In particular, the processing logic can identify (in each the source block and destination block) one or more wordlines that address a set of memory cells (e.g., a page(s)) onto which to program the data referenced by the write command.

[0083] At operation 620, the processing logic programs the write data referenced by the write command to a set of SLC blocks.

[0084] At operation 630, the processing logic selects a set of SLC blocks (e.g., four SLC memory blocks) from the SLC cache to fold (or compact) into a QLC block. The set of memory pages can be selected at random, sequentially, based on a predefined order, based on address location, etc.

[0085] At operation 640, the processing logic performs a data integrity check on the set of SLC blocks. The processing logic can perform the data integrity check on at least a portion of the data from the set of SLC blocks. In some embodiments, the data integrity check can include reading data from a set of sampled memory cells in the set of SLC blocks. In some embodiments, the set of sampled memory cells in the SLC blocks can be one or more memory cells, pages, a wordlines, a group of wordlines in the block, a data portion, or any combination thereof. In some embodiments, the set of sampled memory cells can be selected randomly, can be a predetermined group (e.g., the first, twentieth, and fortieth page of an SLC block), can be a rotating group, etc.

[0086] The data integrity check can verify that the data stored at memory cells does not include any errors, or that the number of errors are below a predetermined threshold. During a scan operation, the processing logic identifies one or more data integrity metrics, such as the bit error count (BEC) or the raw bit error rate (RBER), representing a number of bit errors per unit of time that the data stored at the data block experiences. In some embodiments, during the data integrity check, the processing logic reads a raw code word (i.e., a series of a fixed number of bits) from the page. The processing logic can apply the code word to an error correcting code (ECC) decoder to generate a decoded code word and compare the decoded code word to the raw code word. The processing logic can count a number of flipped bits between the decoded code word and the raw code word, with a ratio of the number of flipped bits to the total number of bits in the code word representing the RBER.

[0087] At operation 650, the processing logic determines whether a value of the data integrity metric (e.g., a BEC value, a RBER value, etc.) satisfies a threshold criterion (e.g., meets or exceeds a threshold value, is lower than a threshold value, etc.). The threshold criterion can be determined and set during manufacturing of the memory sub-system 110 or during programming and/or calibration of the memory sub-system 110. In an example, the processing logic can determine whether an RBER value or a BEC value exceeds a threshold value. If the data integrity metric satisfies a threshold criterion (e.g., BEC or RBER value is above a threshold value), indicating a high error rate associated with data stored on the set of memory pages, the processing logic proceeds to operation 660. If the data integrity metric fails to satisfy the threshold criterion (e.g., BEC or RBER value is below a threshold value), the processing logic proceeds to operation 670, where the processing logic sends, to the memory device, an internal copyback command with respect to the set of SLC blocks.

[0088] At operation 660, the processing logic sends, to the memory device, a corrective copyback command. The corrective copyback command can be used to detect and correct errors in the source data (e.g., the set of SLC blocks), and then write the corrected data to the destination block (e.g., the SLC block). In some embodiments, the corrective copyback command requests the memory device to send the data, from the set of SLC blocks, to the ECC encoder/decoder 111 to correct errors in the data. In particular, the processing logic can perform a scan of the data by reading the data out from the memory device, decoding the source data according to an error correction code (ECC) algorithm, and performing an error check on the source data. In response to detecting that the data contains an error(s), the processing logic can use one or more error handling mechanisms to correct the detected error(s). Error handling mechanisms can be hardware or software based procedures or operations used by the memory sub-system controller to correct an error. Once the data is corrected, the processing logic can send the corrected data to the memory device to be programmed to the destination block. In some embodiments, the processing logic can select an empty block as the destination block. In other embodiments, the processing logic can identify one or more next available wordlines in a block to which to write the source or corrected data. In some embodiments, the destination block can include HLC memory cells. In some embodiments, the destination block can be selected at operation 610. The data can be programmed to the destination block using coarse-fine programming.

[0089] At operation 670, the processing logic sends, to the memory device, an internal copyback command with respect to the source block. More specifically, the processing logic generates an internal copyback command that identifies a set of the source memory cells that store data to be copied to a set of the destination memory cells. The processing logic then transmits the internal copyback command to the control logic (e.g., of the local media controller 135) to be performed by the control logic. Responsive to receiving the internal copyback command, the control logic can initiate a folding operation with respect to the source blocks by enabling the memory device to perform a copyback of the stored data to the set of the destination memory cells. The data can be programmed to the destination memory cells using coarse-fine programming.

[0090] FIG. 7 is a flow diagram of an example method 700 for performing compact management during background maintenance operations, in accordance with some embodiments of the present disclosure. The method 700 can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method 700 is performed by the media management component 113 of FIG. 1A. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

[0091] At operation 710, the processing logic selects an SLC source block. The SLC source block can be selected at random, sequentially, based on a predefined order, based on address location, etc. In some embodiments, the SLC source block is selected from the SLC cache. It is noted that any sized memory granularity (e.g., block, stripe, memory page, etc.) and cell type (e.g., SLC, MLC, TLC, QLC, etc.) can be used.

[0092] At operation 715, the processing logic performs a data validity check on the source block. Data validity can be indicative of a rate of access to the block, the age of the data stored on the block, etc. Data validity can be measured using one or more validity metrics based on program/erase cycles (PECs), time after program (TAP), a block read count, etc. In one example, the data validity check can include obtaining a value associated with a single validity metric (e.g., the amount of PECs that occurred with respect to the block, the TAP of the block, the read count of the block, etc.). In another example, the data validity check can include obtaining a composite value that is a mathematical function (e.g., a weighted sum) of the values of two or more validity metrics.

[0093] A read count refers to using a read counter that tracks the number of read operations performed on each or block, or on any portion of the block (e.g., on a page, on a wordline, etc.). Each time a read operation is performed on a certain block, the memory sub-system controller 115 can increment a corresponding unit level read counter (e.g., block read counter, block read counter, page read counter, wordline read counter, etc.). In some embodiments, the read counters can be maintained in local memory 119. In other embodiments, the read counters can be maintained on memory device 130.

[0094] A PEC refers to a sequence of operations in which data is programmed to a memory cell, then erased from the memory cell, then rewritten to the memory cell. PECs can serve as a metric for quantifying the endurance of a memory cell. To track PEC cycles, the memory sub-system can maintain a PEC counter for each block, or any portion of each block. In some embodiments, the PEC counter can be a write counter that tracks the number of write operations performed on each block (or any portion of each block), an erase counter that tracks the number of erase operations performed on each block, etc. Each time a write operation or erase operation is performed with respect to a certain block,

the memory sub-system controller 115 can increment a corresponding PEC counter. In some embodiments, the PEC counters can be maintained in local memory 119.

[0095] Time After Program (TAP) refers to the time since a memory cell has been written. TAP can be estimated (e.g., inference from a data integrity metric), or directly measured (e.g., from a controller clock). In some embodiments, the memory sub-system controller 115 can maintain a metadata table (in local memory 119) to track the TAP of each block, of each page of a block, of each wordline of a block, etc. In some embodiments, the oldest TAP of a block (e.g., the oldest page programmed in the block) can be used as the TAP value for the entire block. In other embodiments, an average of the TAP for each page can be used as the TAP value for the entire block. In yet other embodiments, other criterion related to the TAP values of any portion or combination of portions of the block can be used as the TAP value for the entire block.

[0096] At operation 720, the processing logic determines whether the data validity value satisfies a threshold criterion. The threshold criterion can include one or more values, one or more conditions, etc. The threshold criterion can be determined and set during manufacturing of the memory sub-system 110 or during programming and/or calibration of the memory sub-system 110. If the data validity metric fails to satisfy (e.g., meets or exceeds, be lower than, etc.) the threshold criterion (e.g., a read count threshold value, a PEC threshold value, and/or TAP threshold value), indicating a relatively low rate of access and/or relatively young data stored on the source block, the processing logic proceeds to operation 725. If the data validity metric satisfies (e.g., meets or exceeds, is lower than, etc.) a threshold criterion (e.g., the read count and/or TAP value is above a threshold value), indicating a high rate of access and/or relatively old data stored on the source block, the processing logic proceeds to operation 730.

[0097] At operation 725, the processing logic selects a new source block. In some embodiments, the new source block is selected from the SLC cache. Responsive to selecting a new source block, the processing logic proceeds to operation 715.

[0098] At operation 730, the processing logic, the processing logic performs a data integrity on the source block. The processing logic can perform the data integrity check on one, some, or all of the pages of the source block. In some embodiments, the data integrity check can include reading data from the set of sampled memory cells in the source block. In some embodiments, the set of sampled memory cells in the source block can be one or more memory cells of the block, a page, a group of pages in the block, a wordline, a group of wordlines in the block, or any combination thereof. In some embodiments, the set of sampled memory cells can be selected randomly, can be a predetermined group (e.g., the first, twentieth, and fortieth memory pages of a block), can be a rotating group, etc.

[0099] The data integrity check can verify that the data stored at memory cells does not include any errors, or that the number of errors are below a predetermined threshold. During a scan operation, the processing logic identifies one or more data integrity metrics, such as the bit error count (BEC) or the raw bit error rate (RBER), representing a number of bit errors per unit of time that the data stored at the data block experiences. In some embodiments, during the data integrity check, the processing logic reads a raw

code word (i.e., a series of a fixed number of bits) from the page. The processing logic can apply the code word to an error correcting code (ECC) decoder to generate a decoded code word and compare the decoded code word to the raw code word. The processing logic can count a number of flipped bits between the decoded code word and the raw code word, with a ratio of the number of flipped bits to the total number of bits in the code word representing the RBER.

[0100] At operation **735**, the processing logic determines whether a value of the data integrity metric (e.g., a BEC value, a RBER value, an amount of memory pages that failed or required error correction procedures, etc.) satisfies a threshold criterion (e.g., meets or exceeds a threshold value, is lower than a threshold value, etc.). The threshold criterion can be determined and set during manufacturing of the memory sub-system **110** or during programming and/or calibration of the memory sub-system **110**. In an example, the processing logic can determine whether an RBER value, a BEC value, a memory page failure value exceeds a threshold value.

[0101] If the data integrity metric fails to satisfy the threshold criterion (e.g., BEC or RBER value is below a threshold value), the processing logic proceeds to operation **725**, where the processing logic selects a new source block. In some embodiments, the new source block is selected from the SLC cache. Responsive to selecting a new source block, the processing logic proceeds to operation **715**.

[0102] If the data integrity metric satisfies a threshold criterion (e.g., BEC, RBER, or memory page failure value is above a threshold value), indicating a high error rate associated with data stored at the block, the processing logic proceeds to operation **740**.

[0103] At operation **740**, the processing logic determines whether additional SLC blocks are available for folding into a QLC block. For example, the processing logic can determine whether the SLC cache contains three additional SLC blocks of data to fill a QLC block. Responsive to determining that the SLC cache include enough SCL blocks (e.g., four or more SLC blocks), the processing logic processes to operation **745** and compacts four SLC blocks into a QLC block. To compact the four SLC blocks, the memory sub-system may perform operations similar to those of method **600** (e.g., determine whether to perform an internal copyback operation or a corrective copyback operation). The additional SLC blocks can be chosen at random, sequentially, based on related data (e.g., as indicated in a translation table, metadata table, data structure, etc.). In some embodiments, the threshold of data integrity check of method **700** (e.g., operation **730** and **735**) can be the same or similar to the threshold of the data integrity check of method **600** (e.g., operation **640** and **650**). Thus, the processing logic can perform the data integrity check of method **600** (e.g., operation **640** and **650**) on the additional SLC blocks only. In other embodiments, the threshold of data integrity check of method **700** (e.g., operation **730** and **735**) can be the different from the threshold of the data integrity check of method **600** (e.g., operation **640** and **650**). Thus, the processing logic can perform the data integrity check of method **600** on all of the SLC blocks of the compaction operation.

[0104] Responsive to determining that the SLC cache does not include enough SCL blocks (e.g., less than four SLC blocks), the processing logic proceeds to operation **750** to reprogram the SLC block to a different location. In some

embodiments, the SLC block can be reprogrammed to refresh the data. For example, once the processing logic reprograms the data on the SLC block to a new SLC block, the read count threshold value, a PEC threshold value, and/or TAP threshold value corresponding to the data is reset. In some embodiments, the processing logic can reprogram the data after performing one or more error corrections operations.

[0105] In some embodiments, the operations of method **700** can be performed as media management operations or maintenance operations. A media management operation can include a garbage collection operation, a wear leveling operation, etc. The media management operation or maintenance operations can be triggered as part of operations to optimize the overall performance of the memory device. For example, the processing logic can enable the optimization operations when the memory sub-system is idle. The source block or block on which the optimization management operation is to be performed can be selected randomly, sequentially, from a list (e.g., a list of blocks containing programmed data, a list of blocks of a particular memory type, such as SLC memory, etc.), in response to a threshold criterion being satisfied (e.g., a parameter associated with the block satisfying a threshold value), etc. In other embodiments, the optimization operation (and the remaining operations discussed with respect to methods **600** and **700**) can be performed on other memory granularities, such as a page, a wordline, a management unit (MU), etc. A MU can be a group of pages across dice and/or channels, and can represent an individual segment of the memory device that can be written or erased in a single operation.

[0106] FIG. **8** is a flow diagram of an example method **800** for performing data compaction in response to a write command, in accordance with some embodiments of the present disclosure. The method **800** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **800** is performed by the media management component **113** of FIG. **1A**. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible. In some embodiments, some operations of method **800** may be similar or the same to those of method **600**.

[0107] The operations of method **800** can relate to performing an internal copyback or a selective copyback of SLC data to QLC data. However, it is noted that the operations of method **800** can be performed between any types of SLC or HLC data (e.g., SLC to HLC, SLC to SLC, HLC to HLC, or HLC to SLC). Method **800** will be discussed in relation to the copyback operations being performed from a set of SLC source blocks to a QLC destination block. However, it should be understood that method **800** can be performed using any sized memory granularity, such as, but not limited to, a set of SLC source

stripes and a HLC destination stripe, a set of SLC source memory pages and a HLC destination memory page, etc.

[0108] At operation **810**, the processing logic receives a write command. The write command can be initiated by a host (e.g., host **120**) or by a memory sub-system controller (e.g., memory sub-system controller **115**). In some embodiments, responsive to receiving the write command, the processing logic can identify a source block(s) in the SLC portion of the memory device (e.g., an SLC cache), and a destination block(s) in a QLC portion of the memory device. In particular, the processing logic can identify (in each the source block and destination block) one or more wordlines that address a set of memory cells (e.g., a page(s)) onto which to program the data referenced by the write command.

[0109] At operation **820**, the processing logic programs the write data referenced by the write command to a set of SLC blocks.

[0110] At operation **830**, the processing logic selects a set of SLC blocks (e.g., four SLC memory blocks) from the SLC cache to fold (or compact) into a QLC block. The set of memory pages can be selected at random, sequentially, based on a predefined order, based on address location, etc.

[0111] At operation **835**, the processing logic determines whether an integrity flag is enabled. The integrity flag can be used to indicate the integrity of the data on each SLC block. If the integrity flag is set, this can indicate that the data on the SLC block may be unreliable, and the processing logic can proceed to operation **840** to perform a data integrity check. If the integrity flag is not set, this can indicate that the data on the SLC block is reliable, and the processing logic can proceed to operation **870** to send, to the memory device, an internal copyback command with respect to the source block.

[0112] The integrity flag can be an indication in a data structure (e.g., a metadata table), a bit, etc. For example, a metadata table can list one or more SLC blocks and, for each SLC block, include integrity flag data (e.g., set to true or false, one or zero, enabled or disabled, etc.). In some embodiments, the integrity flag can be disabled for new SLC block programmed in the SLC cache.

[0113] In some embodiments, the integrity flag can be set (e.g., enabled or disabled) in response to a previous performance of a data integrity check on an SLC block (e.g., operation **730** of method **700**). In some embodiments, the processing logic can perform a data integrity check on one or more of the SLC blocks in the set of SLC blocks and, responsive to one or more SLC blocks satisfying a threshold criterion (e.g., a high reliability error threshold (HREC)), set the integrity flag for the respective SLC block to enabled. In some embodiments, the HREC threshold criterion can be similar or the same to the threshold criterion used in operation **730**. Thus, if the data integrity metric satisfies the threshold criterion of operation **730** (e.g., BEC, RBER, or memory page failure value is above a threshold value), indicating a high error rate associated with data stored at the block, the processing logic can set the integrity flag to enabled. In response to the processing logic performing a relocation operation (e.g., operation **750** of method **700**), the relocation can include the enabled integrity flag for the SLC block. If the data integrity metric fails to satisfy the threshold criterion (e.g., BEC or RBER value is below a threshold value), the processing logic does not set the integrity flag, for the SLC block, to enabled.

[0114] In some embodiments, the HREC threshold for setting the integrity flag can be different from the threshold criterion used in operation **730**. In particular, one BEC value, RBER value, or memory page failure value can be used to trigger operations **725** or **740** of method **700** and a different BEC value, RBER value, or memory page failure value can be used to set the integrity flag. As such, for example, the value from the data integrity check of operation **735** can enable the integrity flag, but be below a threshold value such that the processing logic proceeds to operation **725** to select a new SLC block.

[0115] At operation **840**, the processing logic performs a data integrity check on the set of SLC blocks. The processing logic can perform the data integrity check on at least a portion of the data from the set of SLC blocks. In some embodiments, the data integrity check can include reading data from a set of sampled memory cells in the set of SLC blocks. In some embodiments, the set of sampled memory cells in the SLC blocks can be one or more memory cells, pages, a wordlines, a group of wordlines in the block, a data portion, or any combination thereof. In some embodiments, the set of sampled memory cells can be selected randomly, can be a predetermined group (e.g., the first, twentieth, and fortieth page of an SLC block), can be a rotating group, etc.

[0116] At operation **850**, the processing logic determines whether a value of the data integrity metric (e.g., a BEC value, a RBER value, etc.) satisfies a threshold criterion (e.g., meets or exceeds a threshold value, is lower than a threshold value, etc.). The threshold criterion can be determined and set during manufacturing of the memory sub-system **110** or during programming and/or calibration of the memory sub-system **110**. In an example, the processing logic can determine whether an RBER value or a BEC value exceeds a threshold value. If the data integrity metric satisfies a threshold criterion (e.g., BEC or RBER value is above a threshold value), indicating a high error rate associated with data stored on the set of memory pages, the processing logic proceeds to operation **860**. If the data integrity metric fails to satisfy the threshold criterion (e.g., BEC or RBER value is below a threshold value), the processing logic proceeds to operation **870**, where the processing logic sends, to the memory device, an internal copyback command with respect to the set of SLC blocks.

[0117] At operation **860**, the processing logic sends, to the memory device, a corrective copyback command. The corrective copyback command can be used to detect and correct errors in the source data (e.g., the set of SLC blocks), and then write the corrected data to the destination block (e.g., the SLC block).

[0118] At operation **870**, the processing logic sends, to the memory device, an internal copyback command with respect to the source block. More specifically, the processing logic generates an internal copyback command that identifies a set of the source memory cells that store data to be copied to a set of the destination memory cells. The processing logic then transmits the internal copyback command to the control logic (e.g., of the local media controller **135**) to be performed by the control logic. Responsive to receiving the internal copyback command, the control logic can initiate a folding operation with respect to the source blocks by enabling the memory device to perform a copyback of the stored data to the set of the destination memory cells. The data can be programmed to the destination memory cells using coarse-fine programming.

[0119] FIG. 9 illustrates an example machine of a computer system 900 within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In some embodiments, the computer system 900 can correspond to a host system (e.g., the host system 120 of FIG. 1A) that includes or utilizes a memory sub-system (e.g., the memory sub-system 110 of FIG. 1A) or can be used to perform the operations of a controller (e.g., to execute an operating system to perform operations corresponding to media management component 113 of FIG. 1A). In alternative embodiments, the machine can be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

[0120] The machine can be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0121] The example computer system 900 includes a processing device 902, a main memory 904 (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory 906 (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage system 918, which communicate with each other via a bus 930. Processing device 902 represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device can be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device 902 can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device 902 is configured to execute instructions 926 for performing the operations and steps discussed herein. The computer system 900 can further include a network interface device 908 to communicate over the network 920.

[0122] The data storage system 918 can include a machine-readable storage medium 924 (also known as a computer-readable medium) on which is stored one or more sets of instructions 926 or software embodying any one or more of the methodologies or functions described herein. The instructions 926 can also reside, completely or at least partially, within the main memory 904 and/or within the processing device 902 during execution thereof by the computer system 900, the main memory 904 and the processing device 902 also constituting machine-readable stor-

age media. The machine-readable storage medium 924, data storage system 918, and/or main memory 904 can correspond to the memory sub-system 110 of FIG. 1A.

[0123] In one embodiment, the instructions 926 include instructions to implement functionality corresponding to media management component 113 of FIG. 1A. While the machine-readable storage medium 924 is shown in an example embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0124] Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0125] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. The present disclosure can refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage systems.

[0126] The present disclosure also relates to an apparatus for performing the operations herein. This apparatus can be specially constructed for the intended purposes, or it can include a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program can be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0127] The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems can be used with programs in accordance with the teachings herein, or it can prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these

systems will appear as set forth in the description below. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages can be used to implement the teachings of the disclosure as described herein.

[0128] The present disclosure can be provided as a computer program product, or software, that can include a machine-readable medium having stored thereon instructions, which can be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices, etc.

[0129] In the foregoing specification, embodiments of the disclosure have been described with reference to specific example embodiments thereof. It will be evident that various modifications can be made thereto without departing from the broader spirit and scope of embodiments of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A system comprising:
 - a memory device; and
 - a processing device, operatively coupled to the memory device, to perform operations comprising:
 - performing a data integrity check on a first set of memory cells, configured to store a first number of bits per memory cell, to obtain a data integrity metric value;
 - responsive to determining that the data integrity metric value satisfies a threshold criterion, determining whether a threshold number of second sets of memory cells, configured to store the first number of bits per memory cell, are available for a memory management operation; and
 - responsive to determining that the threshold number of second sets of memory cells are available, causing the memory device to perform the memory management operation by copying data from the first set of memory cells and from the threshold number of second sets of memory cells to a third set of memory cells configured to store a second number of bits per memory cell.
2. The system of claim 1, wherein the operations further comprise:
 - responsive to determining that the data integrity metric value fails to satisfy the threshold criterion, performing a further data integrity check on a third set of memory cells.
3. The system of claim 1, wherein the second number of bits per memory cells is greater than the first number of bits per memory cell.
4. The system of claim 1, wherein the first set of memory cells are configured as single-level cell (SLC) memory.
5. The system of claim 1, wherein the second set of memory cells are configured as higher-level cell (HLC) memory.

6. The system of claim 1, wherein the data integrity metric value reflects at least one of a bit error count (BEC) value or a raw bit error rate (RBER) value.

7. The system of claim 1, wherein the operations further comprise:

- responsive to determining that the threshold number of second sets of memory cells are not available, causing the memory device to copy data from the first set of memory cells and from the threshold number of second sets of memory cells to a fourth set of memory cells configured to store the first number of bits per memory cell.

8. A method, comprising:

- performing a data integrity check on a first set of memory cells, configured to store a first number of bits per memory cell, to obtain a data integrity metric value;
- responsive to determining that the data integrity metric value satisfies a threshold criterion, determining whether a threshold number of second sets of memory cells, configured to store the first number of bits per memory cell, are available for a memory management operation; and

- responsive to determining that the threshold number of second sets of memory cells are available, causing the memory device to perform the memory management operation by copying data from the first set of memory cells and from the threshold number of second sets of memory cells to a third set of memory cells configured to store a second number of bits per memory cell.

9. The method of claim 8, further comprising:

- responsive to determining that the data integrity metric value fails to satisfy the threshold criterion, performing a further data integrity check on a third set of memory cells.

10. The method of claim 8, wherein the second number of bits per memory cells is greater than the first number of bits per memory cell.

11. The method of claim 8, wherein the first set of memory cells are configured as single-level cell (SLC) memory.

12. The method of claim 11, wherein the second set of memory cells are configured as higher-level cell (HLC) memory.

13. The method of claim 8, wherein the data integrity metric value reflects at least one of a bit error count (BEC) value or a raw bit error rate (RBER) value.

14. The method of claim 8, further comprising:

- responsive to determining that the threshold number of second sets of memory cells are not available, causing the memory device to copy data from the first set of memory cells and from the threshold number of second sets of memory cells to a fourth set of memory cells configured to store the first number of bits per memory cell.

15. A non-transitory computer-readable storage medium comprising instructions that, when executed by a processing device operatively coupled to a memory, performs operations comprising:

- performing a data integrity check on a first set of memory cells, configured to store a first number of bits per memory cell, to obtain a data integrity metric value;
- responsive to determining that the data integrity metric value satisfies a threshold criterion, determining whether a threshold number of second sets of memory

cells, configured to store the first number of bits per memory cell, are available for a memory management operation; and

responsive to determining that the threshold number of second sets of memory cells are available, causing the memory device to perform the memory management operation by copying data from the first set of memory cells and from the threshold number of second sets of memory cells to a third set of memory cells configured to store a second number of bits per memory cell.

16. The non-transitory computer-readable storage medium of claim **15**, wherein the operations further comprise:

responsive to determining that the data integrity metric value fails to satisfy the threshold criterion, performing a further data integrity check on a third set of memory cells.

17. The non-transitory computer-readable storage medium of claim **15**, wherein the second number of bits per memory cells is greater than the first number of bits per memory cell.

18. The non-transitory computer-readable storage medium of claim **15**, wherein the first set of memory cells are configured as single-level cell (SLC) memory and the second set of memory cells are configured as higher-level cell (HLC) memory.

19. The non-transitory computer-readable storage medium of claim **15**, wherein the data integrity metric value reflects at least one of a bit error count (BEC) value or a raw bit error rate (RBER) value.

20. The non-transitory computer-readable storage medium of claim **15**, wherein the operations further comprise:

responsive to determining that the threshold number of second sets of memory cells are not available, causing the memory device to copy data from the first set of memory cells and from the threshold number of second sets of memory cells to a fourth set of memory cells configured to store the first number of bits per memory cell.

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